

Climate Models and Interest Rate Risk:

How interest rate risk affects intergenerational equity in climate models

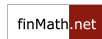
Version 4.2.6

<http://www.christian-fries.de/finmath/intergenerationalequity>

Christian Fries*

LMU

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* The views expressed in this work are the personal views of the authors
and do not necessarily reflect the views or policies of current or previous employers.

Notes

- ▶ Some parts of the slides will be skipped.
DON'T PANIC: It is sufficient to get a basic understanding of the main idea; these slides are available for further reading.
- ▶ Our numerical experiments with the DICE model are available with a graphical user interface at
<https://github.com/finmath/climate-school-exercises/releases>
(check for updates). See also the README at
<https://github.com/finmath/climate-school-exercises>.
- ▶ We provide the full source code for the model and the experiments.

Christian Fries

- ▶ Professor of Computational Insurance and Financial Mathematics (part-time).
- ▶ Head of Model Development at Group Risk Control of a German bank (part-time).
- ▶ Lectures: Numerical Methods, Computational Finance, Applied Mathematical Finance.
- ▶ LMU quantLab: Organization of Workshops and Special Events.
- ▶ Maintaining a large open-source library of numerical methods, [1].
- ▶ Research interests (selection):
 - ▶ Numerical methods: Monte Carlo methods, algorithmic differentiation, ...
 - ▶ Modelling: Interest Rate Models, Hybrid Models, Valuation Adjustments, ...
 - ▶ Implementation: Object-Functional Implementation, Smart Derivative Contracts, High-Performance Computing, ...
- ▶ Book: *Mathematical Finance: Theory, Modeling, Implementation* (Wiley), [4].
- ▶ Current research project related to climate models: [5] with Lennart Quante, *Potsdam Institute for Climate Impact Research (PIK)* (see the last part of this presentation).

Outline

Interest Rates

- Introduction

- Interest Rates

- Compounding

- Valuation (Discounting)

- Relevance in Climate Models

- Another Example

Integrated Assessment Models - Introduction to the DICE Model

- A Short Remark on “Modelling”

- Sketch of the DICE Model

- Abatement Cost

- Economics

- Optimal Emissions Path

Role of Interest Rates in the Model

Implementation / Code Session

- Concepts from Object-Oriented Implementation

- A Java implementation of the DICE Model

Numerical Experiments

- Numerical Experiment: 1-Parameter Abatement Model

- Policy Optimization

- Numerical Experiment: Optimal Emissions Path

Social Cost of Carbon

- Numerical Experiment: Social Cost of Carbon

- The Social Cost of Carbon Does Not Cover the Total Climate Cost

Intergenerational Equity in Integrated Assessment Models

- Numerical Experiment: Distribution of Cost in the One-Parameter Abatement Model

Convexity

Stochastic Interest Rates

IAM with Stochastic Interest Rates

Extensions that Improve Intergenerational Equity

- Adding Funding to Integrated Assessment Models

- Adding Nonlinear Discounting to Integrated Assessment Models

Summary

Source Code and References

Code Sessions and Numerical Experiments

We will discuss the theory and intuition behind **integrated assessment models (IAMs)** - the coupling of a physical climate model with an economic model.

We will then combine the IAM with interest-rate models to analyse the effect of discounting.

A key takeaway is that standard IAMs can exhibit challenges with respect to **intergenerational equity**.

To make these ideas tangible, we will use a **full open-source implementation of the DICE model** together with several interest-rate models and Monte Carlo simulation.

This enables hands-on numerical experiments and, if you like, code exploration.

Three ways to participate:

Lecture mode: simply follow the presentation - nothing required on your side.

Gaming mode: download our demo and run the graphical user interface to play with scenarios and parameters.

Developer mode: import the source code into a development environment (IDE) to inspect, run, and extend the implementation.

Note: Programming experience is not required for lecture or gaming mode. The hands-on part is optional. If you would like to run the code on your own machine, see the optional setup below.

Download the Executable Demos

You can find the numerical experiments (executable and source code)
at <https://github.com/finmath/climate-school-exercises>.

Open-Source Implementation of the Model(s)

You can find the source code of the model implementation in the
package `net.finmath.climate.models.dice`
at <https://github.com/finmath/finmath-lib> (see `src/main/java`).

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INTEREST RATES

INTRODUCTION

Interest Rates

- ▶ Interest rates are the tool to **compare a value or cost at different times**.
- ▶ The link to climate change is evident:
 - ▶ We are facing damages and hence associated costs **in the future**.
 - ▶ We need to finance a transformation in the **present time** to avoid these damages.

Question

- ▶ Why is 100 € now not the same as 100 € in the future?

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Assume you are a bank and lend 100 € to multiple clients for one year. After one year, you request repayment of the 100 € **plus an additional amount**.

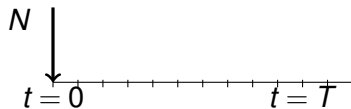
This additional amount is...

- ▶ **Time Value:** A compensation for the fact that you could have used the amount for something else during that time.
- ▶ **Default Premium:** Compensation for the fact that some clients may fail to repay the loan (similar to an insurance premium).

INTEREST RATES

INTEREST RATES

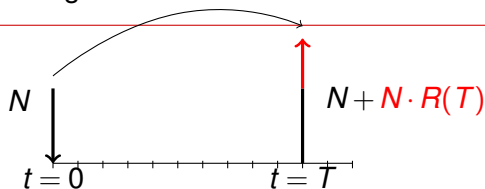
Interest Rates



- If we borrow or lend an amount N today (say at $t = 0$),

investing from time $t = 0$ to $t = T$

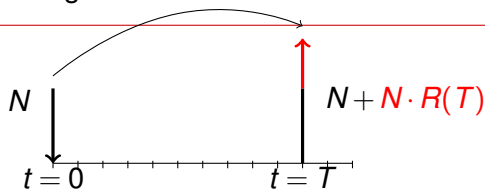
Interest Rates



- If we borrow or lend an amount N today (say at $t = 0$), then we pay or receive back

$$N \cdot (1 + R(T)) = N + N \cdot R(T) \quad \text{at time } T.$$

Interest Rates

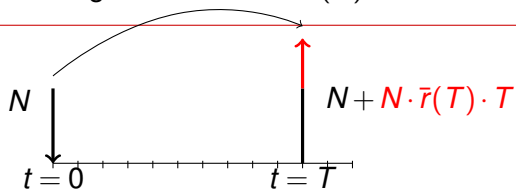


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- The additional amount $N \cdot R(T)$ compensates for the time the money has been borrowed or lent. It depends on the time span $t = 0$ to $t = T$.

Interest Rates

investing at interest rate $\bar{r}(T)$ 

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- The additional amount $N \cdot R(T)$ compensates for the time the money has been borrowed or lent. It depends on the time span $t = 0$ to $t = T$.
- The term **rate** in *interest rate* refers to the fact that this *additional amount* is expressed on a *per-time* basis.
- Expressed on a *per-time* basis, we may define the (linearly compounded) **interest rate**

$$\bar{r}(T) := \frac{1}{T} R(T)$$

INTEREST RATES

COMPOUNDING

Compounding - Exponential Growth

Assume the interest rate for 1 year ($T = t + 1$) is constant: $\bar{r}(t, t + 1) = \bar{r} = \text{const.}$

Let's start at $t = 0$. Lending the amount N_0 for one year, we receive

$$N_0 \cdot (1 + \bar{r}) \quad \text{at } t = 1.$$

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After k years we have

$$N_k := N_0 \cdot (1 + \bar{r})^k \quad \text{at } t = k.$$

The amount grows **exponentially**.

Compounding - Exponential Growth

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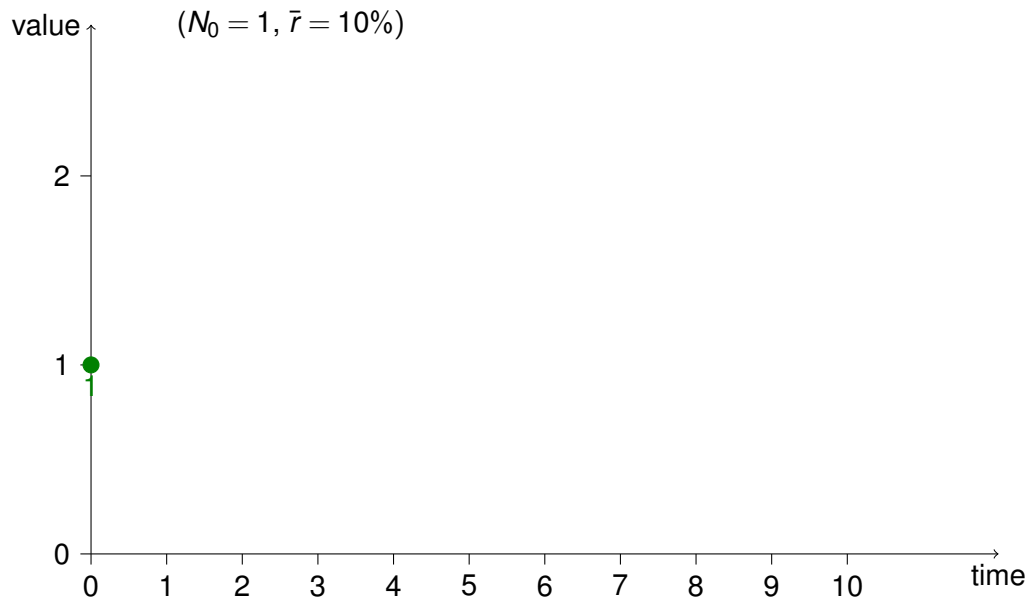
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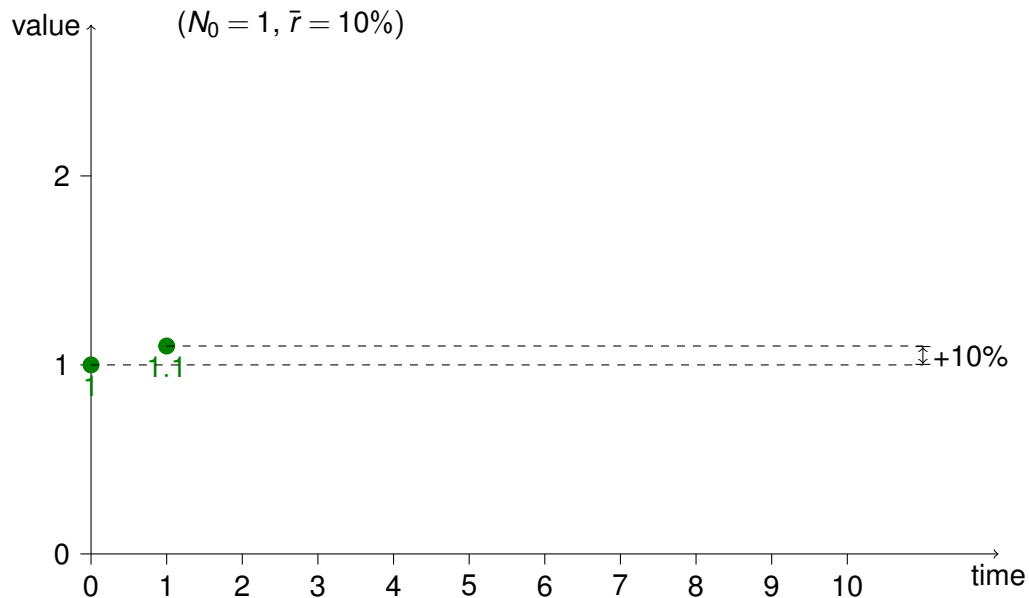
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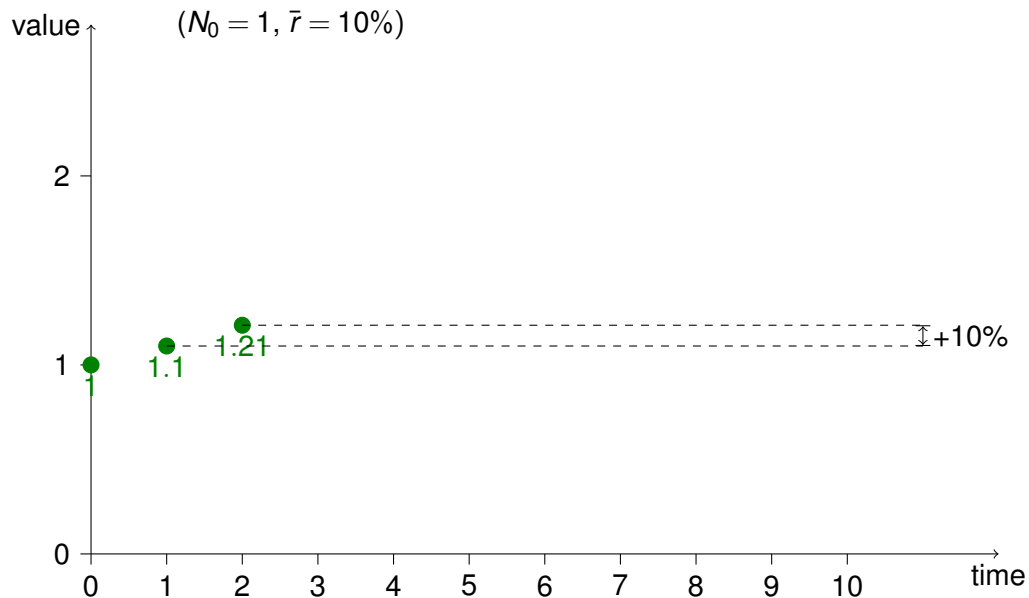
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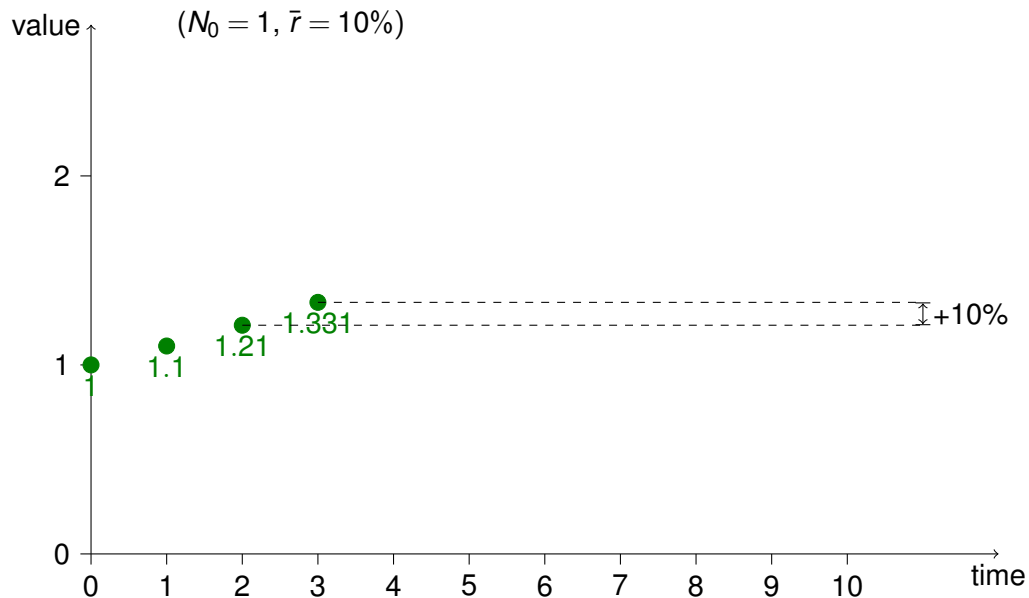
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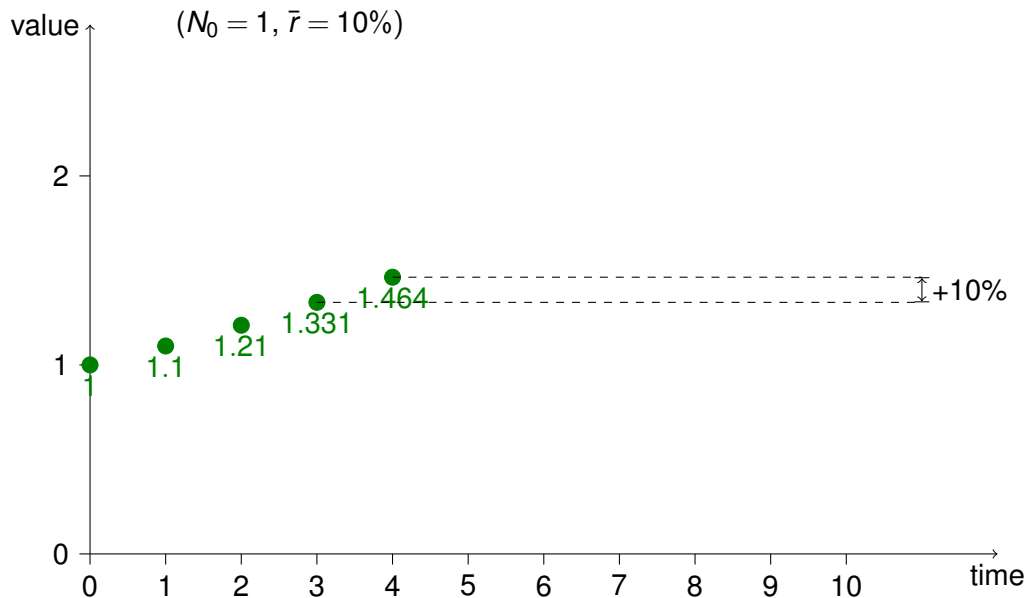
Setting $r = \log(1 + \bar{r})$ we have $N(t) = N_0 \cdot \exp(r \cdot t)$.

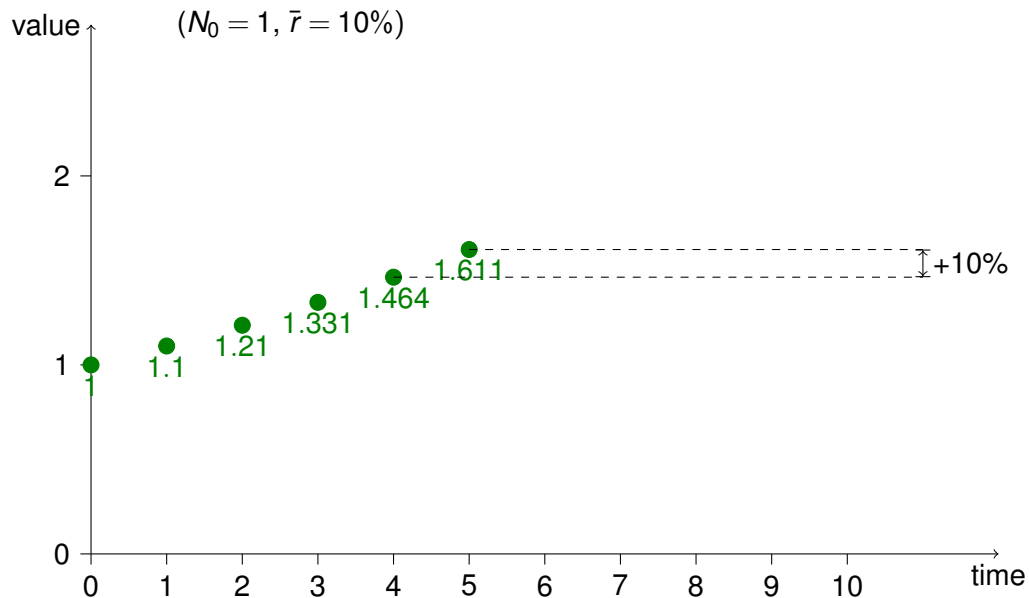


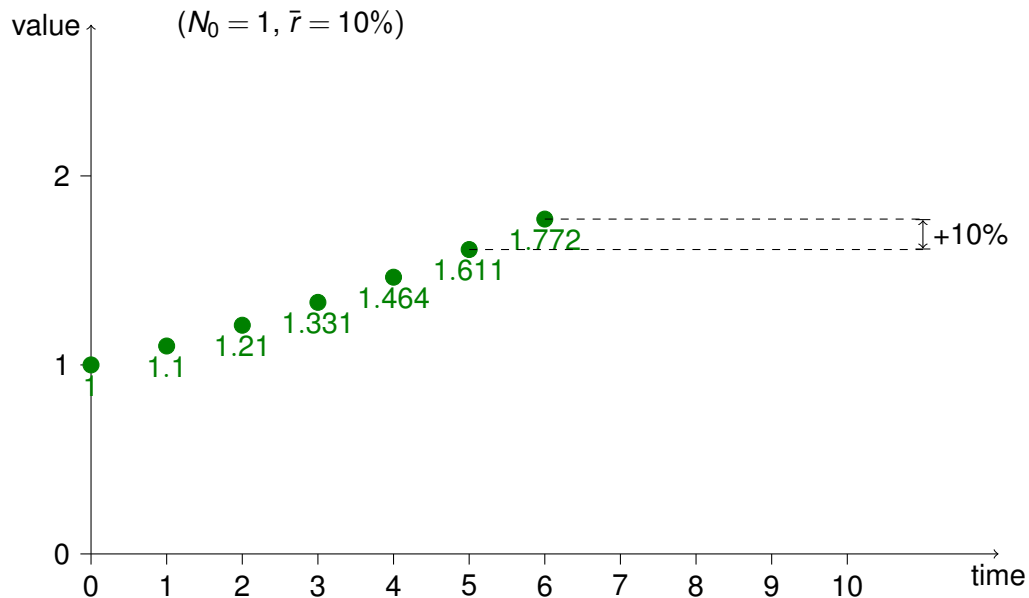


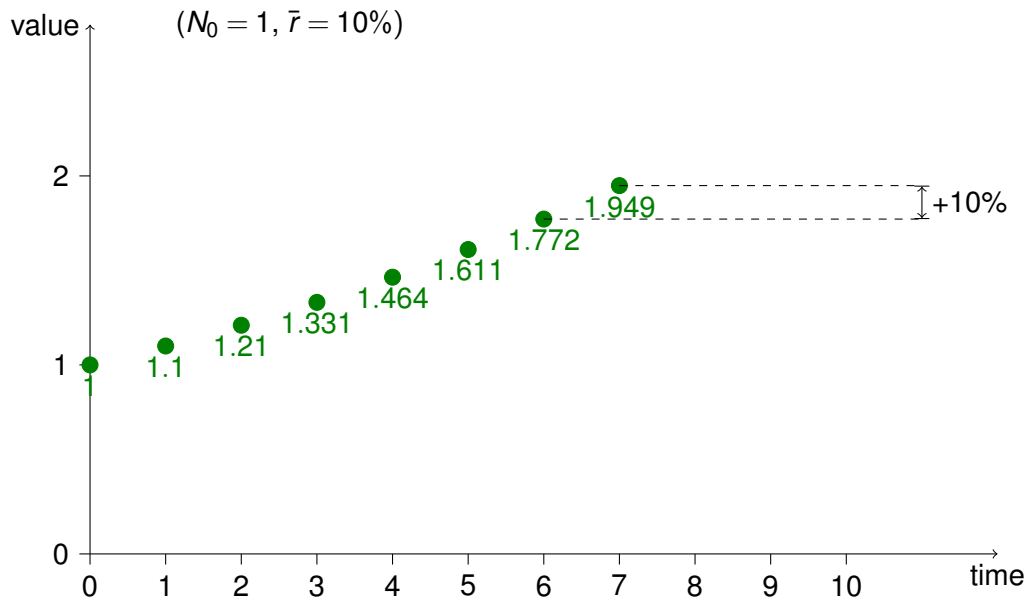


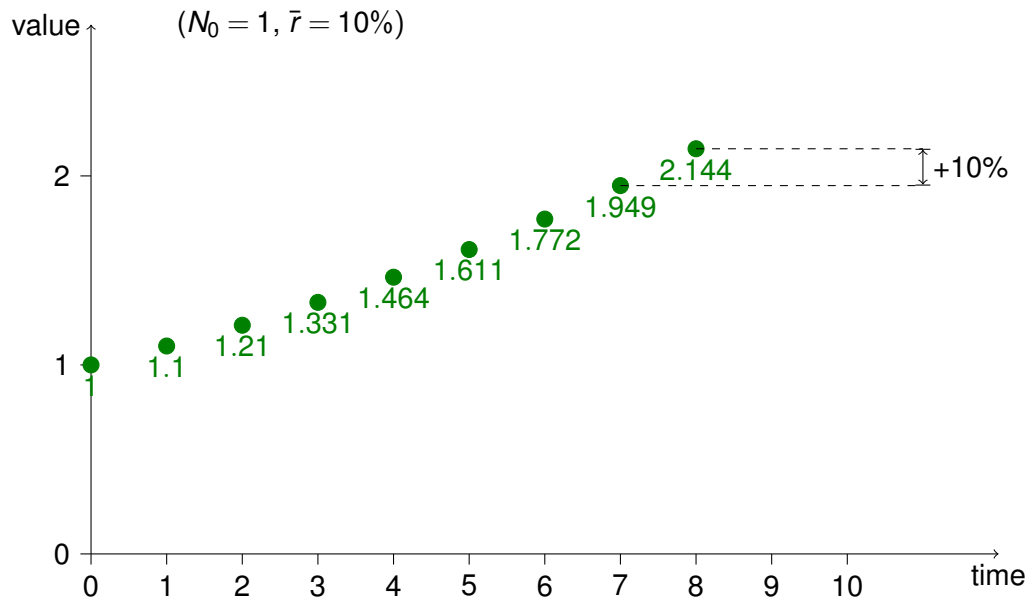


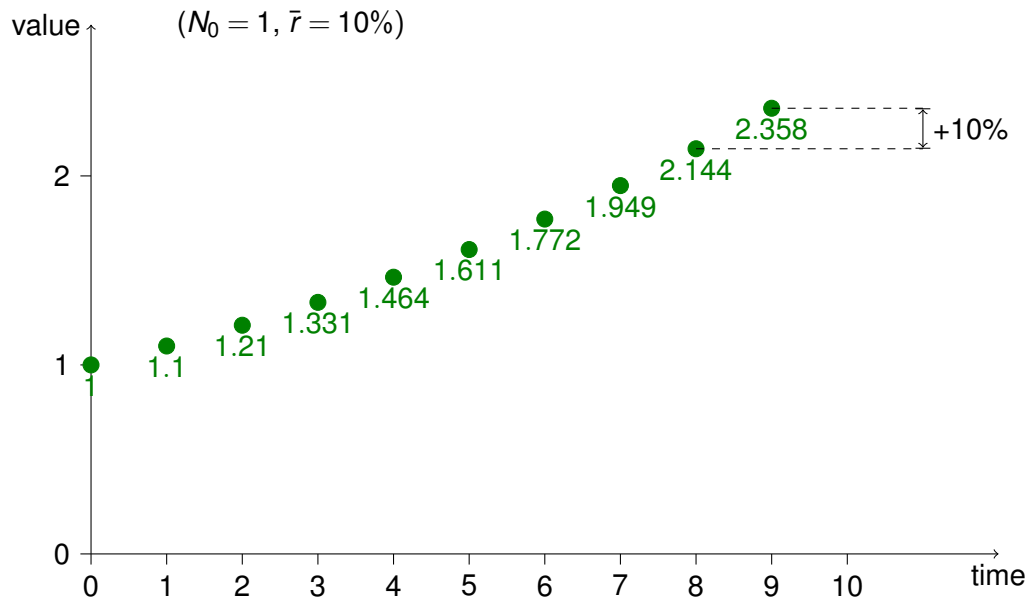


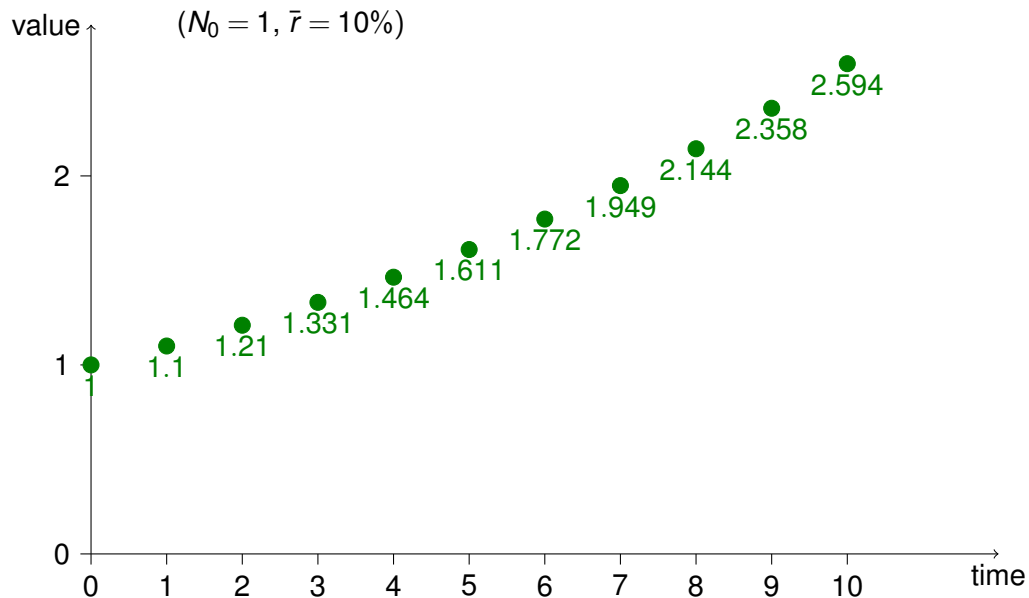


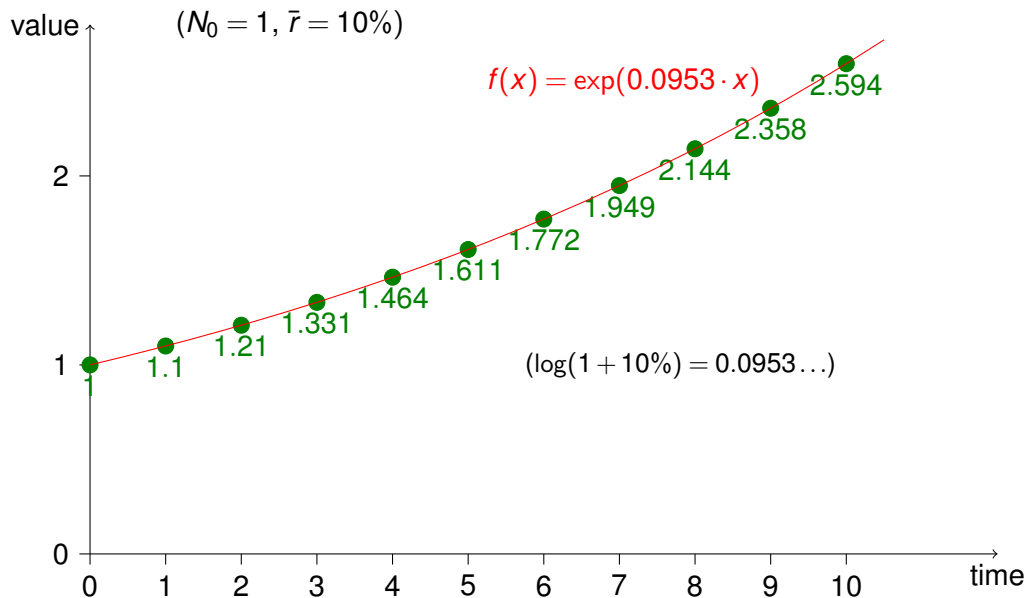












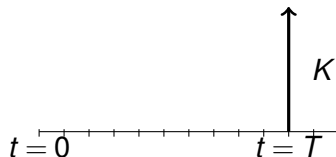
INTEREST RATES

VALUATION (DISCOUNTING)

Valuation

Interest rates tell us how to compare a future **value** to today's **value**.

Why?

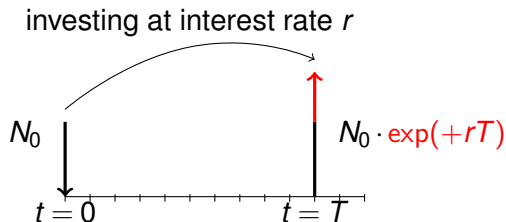


- Consider an amount K , being a payment or liability at the future time T .

Valuation

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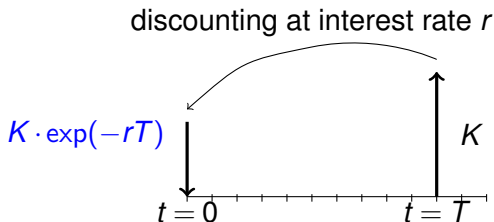
- ▶ Consider an amount K , being a payment or liability at the future time T .
- ▶ If we invest N_0 today, assuming interest rate r , we have after T years

$$N(T) = N_0 \cdot \exp(+r \cdot T)$$

Valuation

Interest rates tell us how to compare a future **value** to today's **value**.

Why?



- ▶ Consider an amount K , being a payment or liability at the future time T .
- ▶ If we invest $N_0 = K \cdot \exp(-r \cdot T)$ today, assuming interest rate r , we have after T years

$$N(T) = N_0 \cdot \exp(+r \cdot T) = K \cdot \exp(-r \cdot T) \cdot \exp(+r \cdot T) = K.$$

- ▶ Consequently:

K at time $t = T$ corresponds to $K \cdot \exp(-r \cdot T)$ at time $t = 0$.

INTEREST RATES

RELEVANCE IN CLIMATE MODELS

Assume there is substantial damage that is considered to cost K at some future time T . Assume that the market borrows and lends money with a compounding interest rate r .

Then (naively) we would value (i.e., *assess*) this damage today ($t = 0$) as

$$K \cdot \exp(-r \cdot T)$$

The factor $\exp(-rT)$ is called a *discount factor*. The valuation is called *discounting*.

Under these stylized assumptions, a future monetary cost K can be matched by investing $K \cdot \exp(-r \cdot T)$ at rate r today.

Damage of size K at time T is valued at $t = 0$ as

$$K \cdot \exp(-rT)$$

For a positive interest rate r , the factor $\exp(-rT)$ is smaller than 1.

It makes the amount K ***much smaller***:

- ▶ For $T = 100$ and $r = 1\%$ we have $\exp(-rT) = 0.37$.
- ▶ For $T = 100$ and $r = 2\%$ we have $\exp(-rT) = 0.14$.
- ▶ For $T = 100$ and $r = 4\%$ we have $\exp(-rT) = 0.02$.

That means, to finance a project requiring 100 million in 100 years, we need to put aside only 14 million now into our bank account (at 2%) or only 2 million (at 4%).

Discounting can make large future monetary costs appear very small today.

It appears as if we can “repair” large damages in the future by very small investments now.

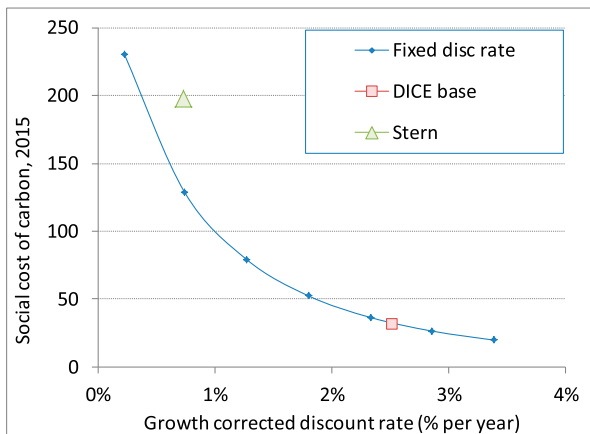


Figure taken from [7]:
Dependence of the *social cost of carbon* (a type of “CO₂ price”) on the discount rate r .

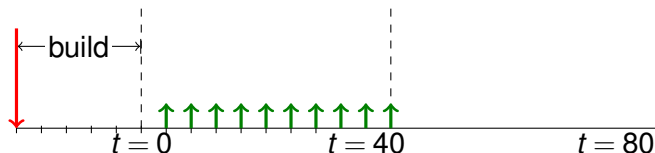
Fig. 3. Social cost of carbon and growth-corrected discount rate in DICE model. The growth-corrected discount rate equals the discount rate on goods minus the growth rate of consumption. The solid line shows the central role of the growth-corrected discount rate on goods in determining the SCC in the DICE model. The square is the SCC from the full DICE model, and the triangle uses the assumptions of *The Stern Review* (10). A further discussion and derivation of the growth-corrected discount rate is given in [Supporting Information](#).

INTEREST RATES

ANOTHER EXAMPLE

Simplified cash flows of a **nuclear power plant**:

- ▶ Very high initial **building cost**.
- ▶ Small operational cost.
- ▶ Smaller **revenue stream** from selling power **for a long time**.

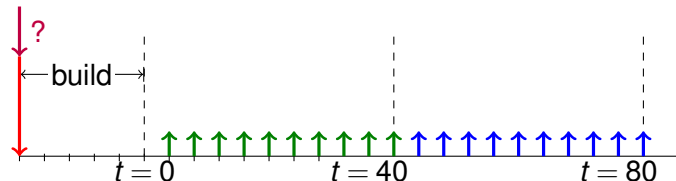


Assumptions:

- ▶ The investor expects a return of 6 % per year on his investment (→ discount rate).
- ▶ The plant runs for **40 years**.
- ▶ The plant is just profitable: Discounted future revenue equals the initial cost.

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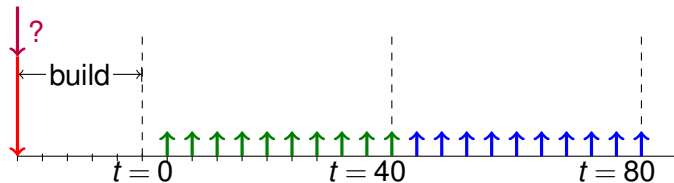
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Question: How much **higher** can the initial construction cost be, if the plant runs **80 years** instead of **40 years**?

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Question: How much **higher** can the initial construction cost be, if the plant runs **80 years** instead of **40 years**?

Answer: 9 %. The additional revenue starts 40 years later and is worth $\exp(-0.06 \cdot 40) \approx 0.0907 \approx 9\%$ of the earlier stream.

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Integrated Assessment Models

are a class of models that **combine**

- ▶ the evolution of climate-related (geophysical) quantities (like emissions and temperature levels) with
- ▶ economic factors (production, consumption).

An optimizing IAM such as DICE is used to find a path that balances reducing emissions (abatement) against reducing future damages: the optimal emission path.

Integrated Assessment Models: DICE Model

- ▶ Nordhaus proposed the DICE model—the **Dynamic Integrated Climate-Economy model**—around 1992. We use the DICE-2016R revision described in [7].
- ▶ We use our generalized implementation [1] and state the corresponding DICE-2016R five-year parametrization. The implementation was checked against the DICE benchmark results. Later experimental changes are stated explicitly.
- ▶ The model is simple, but a good example of the class of *integrated assessment models*. It connects climate and economic quantities—including temperature, carbon amounts, emissions, GDP, consumption, and capital—through calibrated functional forms.
- ▶ Given the DICE model, one tries to find optimal savings-rate and abatement-rate paths that maximize a utility function defined on consumption. This yields an “optimal emissions path”.
- ▶ We give a short introduction to the model (see also [3]).

INTEGRATED ASSESSMENT MODELS - INTRODUCTION TO THE DICE MODEL

A SHORT REMARK ON “MODELLING”

Example: The estimated arrival time from a navigation system:

- ▶ It is not accurate to the second, . . .
- ▶ . . . but accurate enough to guide when we should leave.
- ▶ Varying traffic assumptions or the route gives a useful range of arrival times.

A useful model. . .

- ▶ makes its **assumptions visible**,
- ▶ gives **conditional outcomes**, not promises,
- ▶ can be **tested against observations**, revised and improved.

The alternative to a model is not certainty.

Every decision about the future relies on assumptions. A model makes them **explicit**, **testable** and **revisable**.

The objective is a better decision—not a perfect prediction.

Running a model repeatedly under different assumptions makes uncertainty visible.

Sensitivity analysis: Which assumptions matter?

Vary parameters or model choices and measure the response. This reveals the main drivers of the result.

Scenario analysis: Which outcomes are plausible?

Compare coherent parameter sets—for example, base, best-case and stress scenarios—to obtain a range of **conditional outcomes**. Scenarios need not have probabilities.

Probabilistic or stochastic analysis: What is the **risk**?

Assign probability distributions to uncertain inputs and propagate them through the model. Summarize the output by probabilities, quantiles or prediction intervals—conditional on those input distributions.

Uncertainty becomes an output of the analysis.

Weather: a specific trajectory over a short horizon

What happens at a particular place and time? Weather forecasting is strongly dependent on precise **initial conditions**; small errors grow in a chaotic system.

Climate: a changing distribution over long horizons

How do **averages, ranges and extremes** change as external forcing changes? Climate projections are conditional on emissions scenarios.

- ▶ Not: *Will it rain in Munich on a particular day in 2050?*
- ▶ Rather: **How does an emissions path change the odds of very hot summers?**

Out-of-sample test: published projections versus later observations

- ▶ After accounting for the forcing that actually occurred, **14 of 17 historical climate projections** were consistent with the observed global-mean warming response [6].
- ▶ Ten of the 17 matched observed warming over time directly; this result concerns global mean surface temperature, not every regional impact. x

INTEGRATED ASSESSMENT MODELS - INTRODUCTION TO THE DICE MODEL

SKETCH OF THE DICE MODEL

Time

- ▶ Throughout this presentation, $t = 0$ denotes the DICE-2016R base year 2015.
- ▶ We measure time in years. Quantities that are *per time* (e.g., emissions or consumption) are annualized (i.e., per year).

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- ▶ Example: The total amount of CO₂ added over a time step Δt_i from t_i to t_{i+1} is the integral

$$\int_{t_i}^{t_i + \Delta t_i} E(\tau) d\tau$$

of the emissions $E(t)$ (per time, i.e., annualized emissions).

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- ▶ Example: The total amount of CO₂ added over a time step Δt_i from t_i to t_{i+1} is the integral

$$\int_{t_i}^{t_i + \Delta t_i} E(\tau) d\tau \approx E(t_i) \Delta t_i$$

of the emissions $E(t)$ (per time, i.e., annualized emissions).

This will be approximated by $E(t_i) \Delta t_i$ (sometimes called an *Euler step*).

Time

- ▶ Throughout this presentation, $t = 0$ denotes the DICE-2016R base year 2015.
- ▶ We measure time in years. Quantities that are *per time* (e.g., emissions or consumption) are annualized (i.e., per year).
- ▶ Example: The total amount of CO₂ added over a time step Δt_i from t_i to t_{i+1} is the integral

$$\int_{t_i}^{t_i + \Delta t_i} E(\tau) d\tau \approx E(t_i) \Delta t_i$$

of the emissions $E(t)$ (per time, i.e., annualized emissions).

This will be approximated by $E(t_i) \Delta t_i$ (sometimes called an *Euler step*).

- ▶ We use a time discretization $0 = t_0 < t_1 < \dots < t_n$.
- ▶ The original model uses a fixed time step of $\Delta t_i = 5$ years, and sometimes it may not be clear whether a quantity is an annualized value (such that the total amount over the time step has to be multiplied by Δt_i) or a total amount over the full time step.

Damage

- ▶ The model assumes a **damage function** $\Omega(t)$ describing the fraction of gross output lost because of a rise in atmospheric temperature,

$$\Omega(t) = \phi_1 T_{\text{AT}}(t) + \phi_2 T_{\text{AT}}(t)^2.$$

$\Omega(t)$ is interpreted as a modelled climate-damage loss.

- ▶ The damage function depends on the current **atmospheric temperature change** T_{AT} relative to pre-industrial level (1900).
- ▶ The coefficients of DICE-2016R are $\phi_1 = 0$ and $\phi_2 = 0.00236 \text{ K}^{-2}$.

Simplified

Higher **temperature** leads to more **damage to GDP**.

Temperature

- ▶ The temperature is modelled by a vector in $\mathbb{R}^2$

$$T(t) = \begin{pmatrix} T_{\text{AT}}(t) \\ T_{\text{LO}}(t) \end{pmatrix}$$

where T_{AT} denotes the global surface (atmospheric) temperature and T_{LO} the lower-ocean (deep-ocean) temperature. Here, LO stands for *lower ocean*.

- ▶ Both entries are temperature changes relative to 1900. Their unit is K (Kelvin); a change of 1 K equals a change of 1 °C.
- ▶ The initial temperatures are $T_{\text{AT}}(0) = 0.85$ K and $T_{\text{LO}}(0) = 0.0068$ K.

Temperature Evolution

- ▶ Radiative forcing warms the surface, while heat is exchanged with the deep ocean. The deep ocean reacts more slowly.
- ▶ For a time step Δt_i , our generalized implementation [1] uses

$$T(t_{i+1}) = \Gamma_T(\Delta t_i) \cdot T(t_i) + (\xi_1 F(t_{i+1}), 0)^\top \Delta t_i.$$

The transition matrix $\Gamma_T(\Delta t_i)$ describes the coupling of the two temperature regimes. Here $[F] = \text{W/m}^2$ and $[T] = \text{K}$; ξ_1 converts forcing into temperature change per year.

- ▶ **DICE-2016R parametrization:** $\xi_1 \Delta t_i = 0.1005 \text{ K}/(\text{W/m}^2)$.

Simplified

Higher **radiative forcing** leads to higher **temperature**.

Radiative Forcing

- ▶ The function $F(t)$ is the radiative forcing at time t ($[F] = \text{W/m}^2$).
- ▶ The forcing is a function of the **amount of carbon in the atmosphere** M_{AT} and an external forcing F_{EX} :

$$F(M; t) = F(t) = \eta \log_2 \left(\frac{M_{\text{AT}}(t)}{M_{\text{AT},0}} \right) + F_{\text{EX}}(t),$$

where M_{AT} is the amount of carbon in the atmosphere ($[M] = \text{GtC}$), $M_{\text{AT},0} = 588 \text{ GtC}$ and $\eta = 3.6813 \text{ W/m}^2$. The external forcing $F_{\text{EX}}(t)$ rises linearly from 0.5 W/m^2 in 2015 to 1.0 W/m^2 in 2100 and is constant thereafter.

- ▶ See https://en.wikipedia.org/wiki/Radiative_forcing.

Simplified

A higher **amount of carbon in the atmosphere** leads to higher **radiative forcing**.

Forcing (temperature change)

- ▶ The parameter $M_{AT,0} = 588$ GtC is the pre-industrial equilibrium amount of atmospheric carbon used by DICE-2016R.
- ▶ Note: $1 \text{ ppm CO}_2 \hat{=} 2.12 \text{ GtC} = 7.76 \text{ GtCO}_2$, i.e. $588 \text{ GtC} \hat{=} 277 \text{ ppm CO}_2$.

Amounts of Carbon

- ▶ DICE tracks the amount of carbon in three reservoirs:

$$M(t) = \begin{pmatrix} M_{\text{AT}}(t) \\ M_{\text{UO}}(t) \\ M_{\text{LO}}(t) \end{pmatrix},$$

where M_{AT} is the amount in the atmosphere, M_{UO} the amount in the upper ocean, and M_{LO} the amount in the lower ocean.

- ▶ The unit of an element of the vector is $[M.] = \text{GtC}$ (gigatonnes of carbon).

Evolution of Carbon Amounts

- ▶ Emissions add carbon to the atmosphere. Over time, some carbon moves between the atmosphere, the upper ocean, and the lower ocean.
- ▶ For a time step Δt_i , our generalized implementation [1] uses

$$M(t_{i+1}) = \Gamma_M(\Delta t_i) \cdot M(t_i) + \left(\frac{E(t_i)\Delta t_i}{3.666}, 0, 0 \right)^\top.$$

The transition matrix $\Gamma_M(\Delta t_i)$ describes the coupling of the three carbon reservoirs. Here $[E] = \text{GtCO}_2/\text{year}$ and $[M] = \text{GtC}$; the factor $1/3.666$ converts CO_2 to carbon.

- ▶ **DICE-2016R parametrization:** the atmospheric input is $5E(t_i)/3.666$.
- ▶ This is the *carbon cycle*, modelling the exchange between the atmosphere, land, and ocean—see https://en.wikipedia.org/wiki/Carbon_cycle.

Simplified

Emissions increase the **amount of carbon in the atmosphere**.

Emissions

- ▶ The emissions function $E(t)$ gives the annual emissions rate ($[E] = \text{GtCO}_2/\text{year}$).
- ▶ Emissions are split into two components: industrial emissions (proportional to GDP) E_{GDP} and external emissions E_{EX} (e.g., land use),

$$E(t) = E_{\text{GDP}}(t) + E_{\text{EX}}(t).$$

Industrial Emissions

- ▶ Industrial emissions are given by a linear function of GDP:

$$E_{\text{GDP}}(t) = \sigma(t) \cdot (1 - \mu(t)) \cdot \text{GDP}(t).$$

Here $\text{GDP}(t)$ is annualized GDP, with $[\text{GDP}] = 10^{12}\text{USD}/\text{year}$ and $[E_{\text{GDP}}(t)] = \text{GtCO}_2/\text{year}$.

- ▶ In these slides, $\text{GDP}(t)$ denotes gross output. Original DICE distinguishes gross output Y_{GROSS} from output after damages and abatement, denoted by Y .
- ▶ The function μ is the **abatement rate**. It describes the fraction of industrial emissions that is abated. Increasing μ creates costs (see below).
- ▶ The function σ models the emission intensity, $[\sigma(t)] = \text{GtCO}_2/10^{12}\text{USD}$, see below.

Simplified

Higher **GDP** produces more industrial **emissions**, reduced by the **abatement** factor μ .

External Emissions

- ▶ The external emissions $E_{\text{EX}}(t)$ are an annual emissions rate ($[E_{\text{EX}}] = \text{GtCO}_2/\text{year}$).

Total Emissions

- ▶ The total emissions are the sum of the industrial and external emissions:

$$E(t) = E_{\text{GDP}}(t) + E_{\text{EX}}(t)$$

Emission intensity σ is the amount of industrial CO₂ emitted per unit of economic output, with $[\sigma] = \text{GtCO}_2 / (10^{12} \text{ USD})$. It declines over time, independently of deliberate abatement μ .

- For a time step Δt_i , our generalized implementation [1] uses

$$\begin{aligned}\delta_\sigma(t) &= \delta_{\sigma,0} \exp(-d \cdot t), \\ \sigma(t_{i+1}) &= \sigma(t_i) \exp(-\delta_\sigma(t_i) \Delta t_i) .\end{aligned}$$

- **DICE-2016R parametrization:** $\delta_{\sigma,0} = 0.0152/\text{year}$ and $d = -\log(1 - 0.001)/\text{year}$.
- The implementation stores the correspondingly rescaled intensity 35.85/105.5 and applies the factor $1/(1 - \mu(0))$ separately. With $\mu(0) = 0.03$, this gives the DICE initial intensity $35.85/[105.5(1 - 0.03)]$.

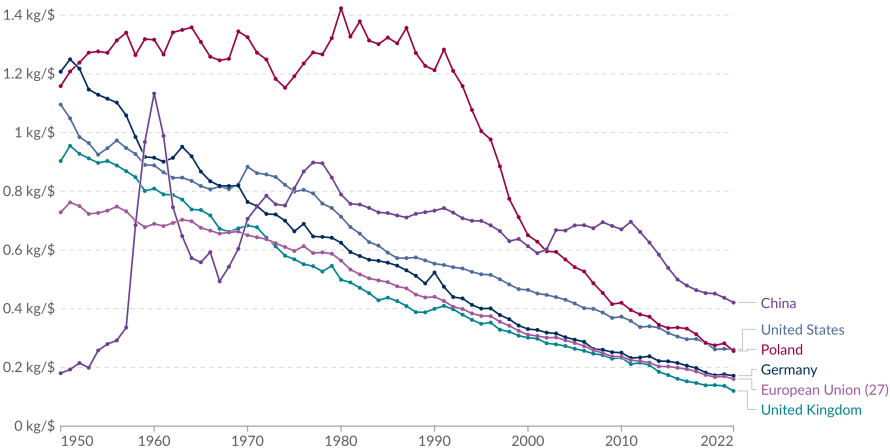
Simplified

Over time, each unit of economic output produces less CO₂ even before additional abatement.

Carbon intensity: CO₂ emissions per dollar of GDP



Kilograms of CO₂ emitted per dollar of GDP. Fossil fuel and industry emissions are included. Land-use change emissions are not included. GDP data is adjusted for inflation and differences in living costs between countries.

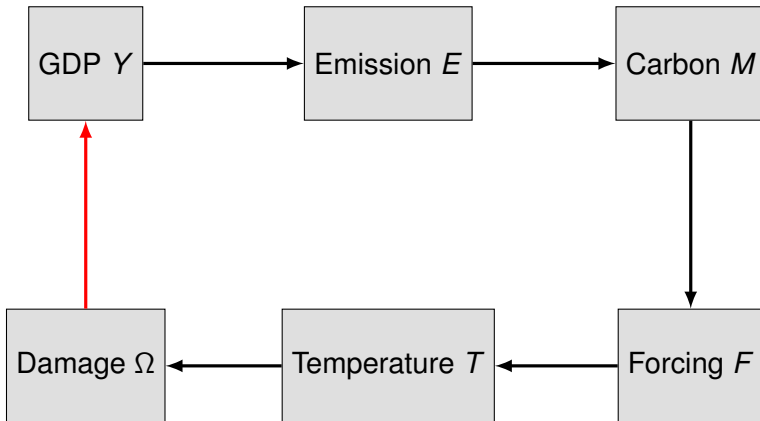


Data source: Global Carbon Budget (2024); Bolt and van Zanden – Maddison Project Database 2023

Note: GDP data is expressed in international-\$ at 2011 prices.

OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

At this point, we see that emissions - as a function of the GDP - induce damage to the GDP.



INTEGRATED ASSESSMENT MODELS - INTRODUCTION TO THE DICE MODEL

ABATEMENT COST

Abatement Costs / Backstop Price

- ▶ The parameter $\mu(t)$ defines the fraction of emissions we are reducing (abating).
- ▶ The *abatement cost* function is a cost per unit of potential industrial emissions and is given by

$$\Lambda(t) = B(t) \cdot \frac{1}{\theta_2} \mu(t)^{\theta_2},$$

where $B(t)$ is the *backstop price* and is modelled as

$$B(t) = B(0) \cdot \exp(-b \cdot t),$$

with $B(0) = 550.0/1000.0 \cdot 10^{12}$ USD/GtCO₂ and $b = -\log(1 - 0.025)/5$ and $\theta_2 = 2.6$.

- ▶ The unit of Λ is $[\Lambda] = 10^{12}\text{USD/GtCO}_2 = 1000\text{USD/tCO}_2$.

Simplified

The larger the **abatement** $\mu(t)$, the larger the **abatement cost** factor $\Lambda(t)$.

Abatement Costs

- ▶ Abatement costs apply to potential industrial emissions before abatement. They are

$$C_A(t) := \Lambda(t) \cdot \sigma(t) \cdot GDP(t).$$

- ▶ The unit of C_A is 10^{12} USD/year (trillion USD per year).
- ▶ In the original model, the factor corresponding to Λ includes σ and is multiplied directly by GDP .
- ▶ Note that these costs apply every year, just as GDP is measured per year.

Abatement Costs: Interpreting the Abatement Cost Model

- ▶ The abatement cost factor is given by the function $\mu \mapsto \Lambda(\mu, B, \theta_2) = B \cdot \frac{1}{\theta_2} \mu^{\theta_2}$:
 - ▶ For $\mu = 0$ we have $\Lambda(0, B, \theta_2) = 0$.
 - ▶ For $\mu = 1$ we have $\Lambda(1, B, \theta_2) = B/\theta_2$ and $\left(\frac{\partial}{\partial \mu} \Lambda\right)(1, B, \theta_2) = B$.
- ▶ In other words:
 - ▶ $B(t)/\theta_2$ is the average cost per abated unit when abatement reaches 100%.
 - ▶ $B(t)$ is the marginal cost of abating the last unit of tCO₂ ($\mu(t)$ close to 1).
- ▶ At $t = 0$, this gives
 - ▶ For $\mu = 1$, the average price is $550.0/2.6 \text{ USD/tCO}_2 \approx 212 \text{ USD/tCO}_2$.
 - ▶ For $\mu = 1$, the marginal price for the last unit is 550.0 USD/tCO_2 , if abatement is performed now.
- ▶ The choice of b will recover the value of the original model, which is

$$\exp(-b \cdot t) = (1 - 0.025)^{t/5}$$

which is given by the assumption of a 2.5 % price reduction every 5 years.

INTEGRATED ASSESSMENT MODELS - INTRODUCTION TO THE DICE MODEL

ECONOMICS

Net GDP (Remaining GDP) and Savings Rate

- ▶ A fraction $\Omega(t)$ of gross output is lost to modelled climate damage.
- ▶ $\Lambda(t) \cdot \sigma(t) \cdot GDP(t)$ is removed from GDP to pay for abatement (reduction of emissions).
- ▶ The remainder

$$GDP_{\text{net}} := \underbrace{Y(t)}_{\text{GDP}} - \underbrace{\Omega(t)Y(t)}_{\text{Damage Cost } C_D} - \underbrace{\Lambda(t) \cdot \sigma(t) \cdot GDP(t)}_{\text{Abatement Cost } C_A},$$

may be used for **investment** $I(t)$ or **consumption** $C(t)$, where the allocation between the two is defined by the savings rate $s(t)$,

$$\begin{aligned} I(t) &:= s(t) GDP_{\text{net}}, \\ C(t) &:= (1 - s(t)) GDP_{\text{net}}. \end{aligned}$$

The savings rate function s is (besides the abatement μ) the second free parameter that is used in the (policy) optimization.

For a time step Δt_i , our generalized implementation [1] uses

$$K(t_{i+1}) = \underbrace{\exp(-\delta_K \Delta t_i) K(t_i)}_{\text{depreciated capital}} + \underbrace{I(t_i) \Delta t_i}_{\text{investment}}.$$

DICE-2016R parametrization: $\delta_K = -\log(0.9)/\text{year}$, giving $K(t_{i+1}) = 0.9^5 K(t_i) + 5I(t_i)$. Its inequality form binds in the standard solution.

Capital, productivity A , and population L determine gross output:

$$Y(t) = A(t)K(t)^\gamma \left(\frac{L(t)}{1000} \right)^{1-\gamma}.$$

Simplified

After depreciation, **investment** is added to **capital**. Together with productivity and population, capital helps determine the next period's **GDP**.

Utility / Welfare

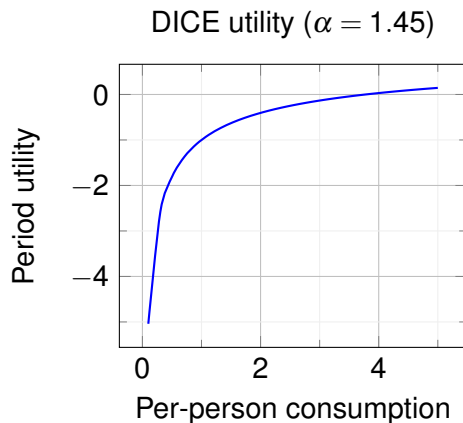
For per-person consumption $c(t) = 1000C(t)/L(t)$ (in thousand USD per person per year), our generalized implementation [1] defines period welfare by

$$V(t) = U(c(t), L(t)) = L(t) \left(\frac{c(t)^{1-\alpha} - 1}{1-\alpha} - 1 \right)$$

- ▶ **DICE-2016R parametrization:** $\alpha = 1.45$.
- ▶ An additional dollar raises welfare, but matters less when consumption is already high.

Simplified

Consumption determines the the **utility** and thus the period **welfare**.

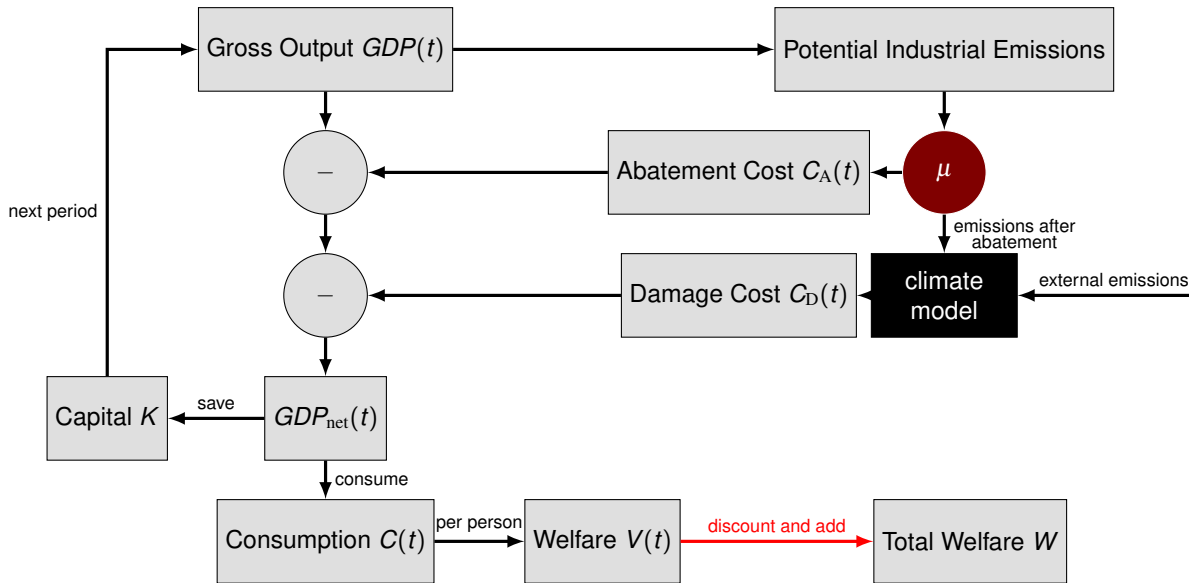


Total social welfare is obtained by adding discounted period welfare. In our generalized implementation [1],

$$W(r) = \sum_i V(t_i) \cdot \exp(-r \cdot t_i) \Delta t_i.$$

DICE-2016R parametrization: $r = \log(1 + 0.015)/\text{year}$. DICE-2016R applies an additional positive affine scaling, which changes the reported welfare value but not the optimizer. Later experiments deliberately vary r .

The model chooses the abatement path $\mu(t)$ and savings-rate path $s(t)$ to maximize total welfare: μ balances abatement costs against climate damage, while s balances consumption against investment.



INTEGRATED ASSESSMENT MODELS - INTRODUCTION TO THE DICE MODEL

OPTIMAL EMISSIONS PATH

Policy Optimization—Optimal Emissions Path

- ▶ **Policy optimization** tries to maximize total welfare by choosing the abatement function $t \mapsto \mu(t)$ and the savings-rate function $t \mapsto s(t)$.
- ▶ The optimal functions $t \mapsto \mu(t)$ and $t \mapsto s(t)$ may depend on other quantities, e.g., the interest rate r .

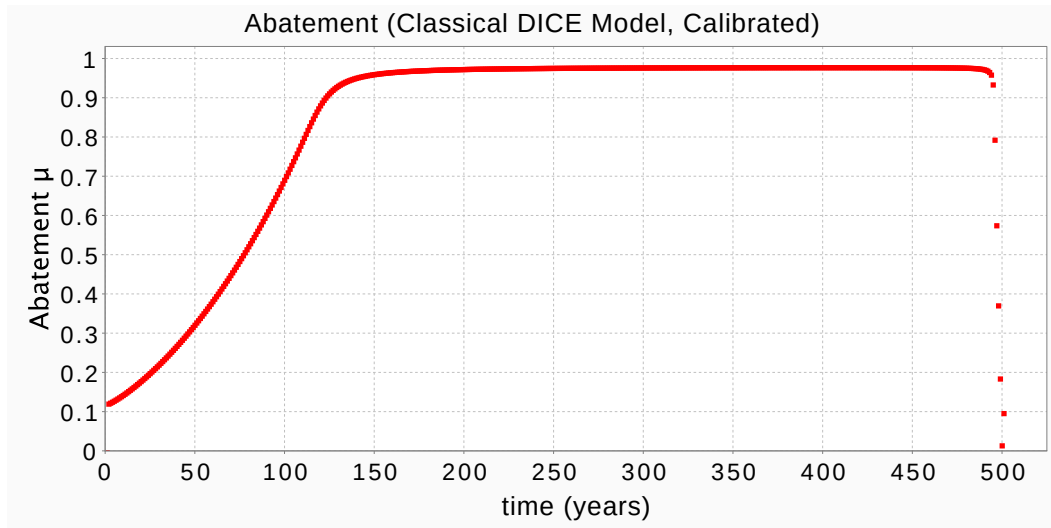


Figure 1: The calibrated (optimal) abatement function $t \mapsto \mu(t)$ for our finite-horizon DICE implementation.

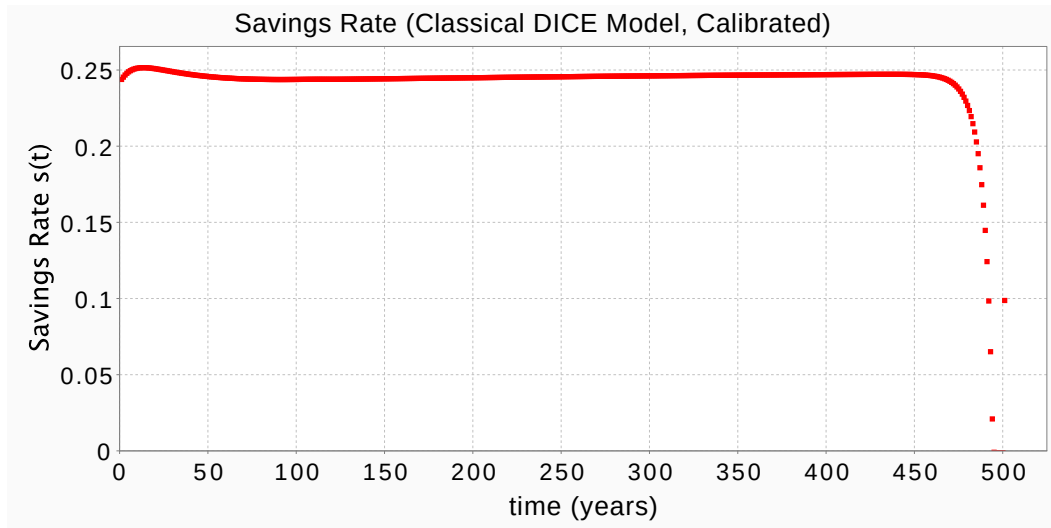


Figure 2: The calibrated (optimal) savings rate function $t \mapsto s(t)$ for our finite-horizon DICE implementation.

Policy Optimization

Question: Why does our policy optimization generate the drop to zero in the optimal abatement and savings rates near $t = 500$?

Policy Optimization

Question: Why does our policy optimization generate the drop to zero in the optimal abatement and savings rates near $t = 500$?

Answer: The model has a finite time horizon. This means utility and welfare beyond $t = 500$ are not considered. In this implementation, it is optimal to stop all abatement and savings, and consume everything before the model horizon ends.

Note: The original DICE model mitigates the terminal savings artifact by fixing the final ten savings rates to a long-run value; our unconstrained implementation does not. Extending the horizon does not alter the result significantly except near the new endpoint, although this must be verified.

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Interest Rates in the DICE Model

- ▶ **Our generalized implementation** [1]: period welfare is discounted with $\exp(-rt)$.
- ▶ **DICE-2016R parametrization:** $r = \log(1 + 0.015)/\text{year}$. The 1.5% is the pure rate of social time preference; the model's return on goods is a different, endogenous quantity.
- ▶ In the following numerical experiments, we deliberately vary r as a scenario parameter.
- ▶ A higher r gives distant damages less weight and tends to postpone abatement in this experiment.

Dependence of Optimized $T^{\mu=1}$ on the Discount Rate r

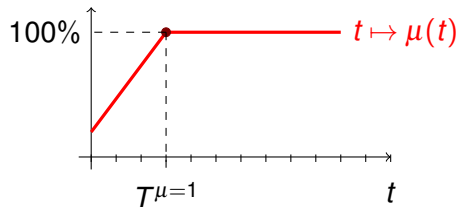
Consider the simple one-parameter abatement model:

$$\mu(t) = \min \left(\mu(0) + \frac{1 - \mu(0)}{T^{\mu=1}} t, 1.0 \right),$$

where the parameter $T^{\mu=1}$ represents the time to reach 100% abatement of modelled industrial emissions.

Interpretation: The time $T^{\mu=1}$ is when modelled industrial emissions reach 100% abatement. It is the only free parameter.

The abatement policy is piecewise linear.



The following figure shows how the optimized $T^{\mu=1}$ depends on the discount-rate level:

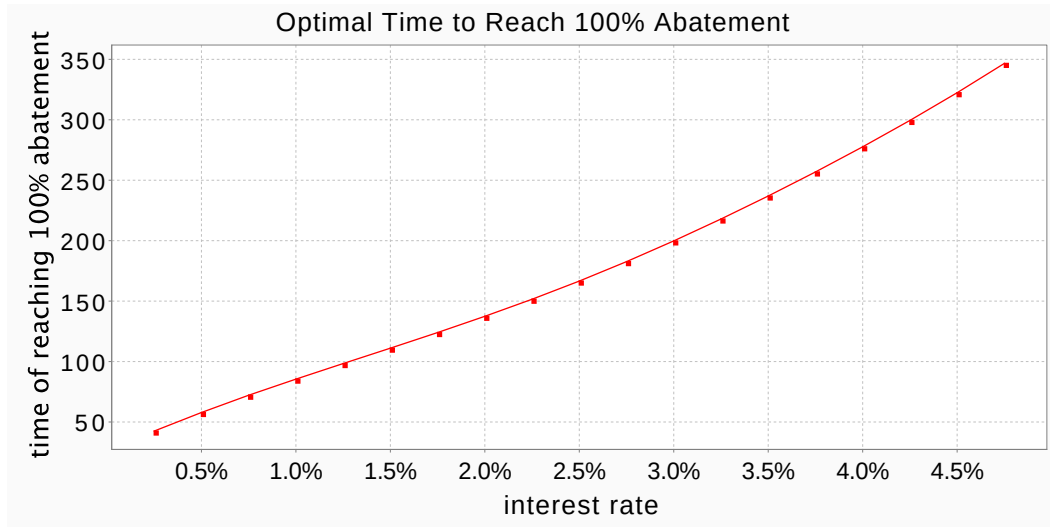


Figure 3: The dependence of $T^{\mu=1}$ on the discount-rate level (for an optimized constant savings rate s_0).

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Implementation / Code Session

- ▶ We review a Java implementation of the DICE model and its policy optimization.
- ▶ We will briefly review all the components of our DICE model. They are represented in a modular way by *interfaces* and *classes* (e.g., for the damage function or carbon cycle). The corresponding *objects* are then combined to form the model.
- ▶ For those unfamiliar with programming, we briefly explain the concepts.
- ▶ Based on our code we will conduct several experiments. We try to enable you to run such experiments on your own machine.
- ▶ Note: It is not required to have an in-depth understanding of our code to run experiments with the model.

IMPLEMENTATION / CODE SESSION

CONCEPTS FROM OBJECT-ORIENTED IMPLEMENTATION

Concepts from Object-Oriented Implementation (Java) (Short)

The main components are

- ▶ **interface** - a collection of functions (sometimes called *methods*) that describe **what can be done** with an *object* of a *class*.
Example: A damage function provides the damage for a given temperature.
- ▶ **class** - the implementation of an interface, i.e., the specification of **how it is done** and what kind of **data** is associated with the corresponding *object*.
- ▶ **object** - an instance of a *class* with concrete **data** (if any) that can be manipulated by the algorithm specified in the `class`. It is created (instantiated) with the keyword **new**.

In addition, there are other helpful constructs, e.g.,

- ▶ **package** - a name (prefix) for classes that organizes them. This is similar to a folder (directory) that organizes files.

IMPLEMENTATION / CODE SESSION

A JAVA IMPLEMENTATION OF THE DICE MODEL

A Java Implementation of the DICE Model (for Research Purposes)

The following code can be found under the

package `net.finmath.climate.models.dice`

in the repository `https://github.com/finmath/finmath-lib`

- ▶ The **package** `net.finmath.climate.models.dice` contains ...
 - ▶ the **class** `DICEModel`.

- ▶ The **package** `net.finmath.climate.models.dice.submodels` contains ...
 - ▶ the *classes*
 - ▶ `AbatementCostFunction`,
 - ▶ `DamageFromTemperature`,
 - ▶ `EvolutionOfCarbonConcentration`,
 - ▶ `EvolutionOfTemperature`,
 - ▶ etc.

Time

```
interface TimeDiscretization in package net.finmath.time
```

```
class TimeDiscretizationFromArray
```

Random Variable

interface `RandomVariable` in **package** `net.finmath.stochastic`

- ▶ A standard DICE run has no stochastic state variables. Uncertainty remains across parameterizations, scenarios, and model structures.
- ▶ However, our implementation allows quantities to become stochastic.
- ▶ This ability is achieved by using the interface `RandomVariable`.
- ▶ For the standard DICE model, `RandomVariable` is represented by a deterministic implementation (e.g., `Scalar`).
- ▶ We have an extended version of the DICE model with a stochastic interest-rate model (and a stochastic abatement model).

Damage

```
interface DoubleUnaryOperator
```

```
class DamageFromTemperature
```

Temperature

```
interface Temperature
```

```
class Temperature2DScalar
```

Temperature Evolution

```
class EvolutionOfTemperature
```

Radiative Forcing (as a function of the amount of atmospheric carbon)

```
interface BiFunction
```

```
class ForcingFunction
```

```
interface Function
```

```
class ForcingExternalFunction
```

Carbon Amounts

```
interface CarbonConcentration
```

```
class CarbonConcentration3DScalar
```

Carbon Evolution

```
interface TriFunction
```

```
class EvolutionOfCarbonConcentration
```

Emission

```
interface BiFunction
```

```
class EvolutionOfEmissionIndustrialIntensity
```

Abatement Cost

```
interface BiFunction
```

```
class AbatementCostFunction
```

Evolution of Capital

```
interface Function
```

```
interface BiFunction
```

```
class EvolutionOfCapital
```

Other Elements

- ▶ Some other elements are represented by simple floating-point numbers, e.g., current GDP.
- ▶ In the simple DICE model the interest rate is a constant r .
- ▶ We will later add a stochastic interest-rate model.

Abatement Model and Savings Rate Model - the free parameters

```
interface AbatementModel
```

```
interface SavingsRateModel
```

DICE Model and Policy Optimization

```
interface ClimateModel
```

```
class DICEModel
```

The class `DICEModel`

- ▶ consumes an abatement model (fixed abatement function) and a savings-rate model (fixed savings-rate function), ...
- ▶ ... glues together all the other model components, ...
- ▶ ... and calculates the social welfare.

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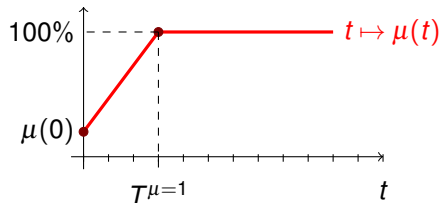
NUMERICAL EXPERIMENTS

NUMERICAL EXPERIMENT: 1-PARAMETER ABATEMENT MODEL

Numerical Experiment: One-Parameter Abatement Model

- ▶ Instantiate an object of `class` `DICEModel`.
 - ▶ Use a constant savings rate $t \mapsto 0.26$.
 - ▶ Use a simple one-parameter abatement model, with the single parameter $T^{\mu=1}$: linear from 0.03 at $t = 0$ to 1.0 at $t = T^{\mu=1}$, then constant at 1.0.

$$\mu(t) = \min \left(\mu(0) + \frac{1 - \mu(0)}{T^{\mu=1}} t, 1.0 \right),$$

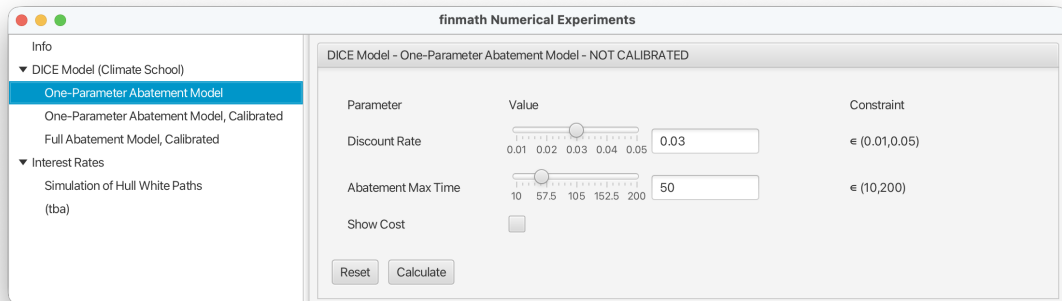


- ▶ Extract from the model the calculated values of
 - ▶ the evolution of atmospheric carbon,
 - ▶ the temperature evolution,
 - ▶ etc.
- ▶ Explore the results for different choices of $T^{\mu=1}$ (e.g., 50, 100) and the interest rate r .

Note: We provide some helper code to plot these quantities for a given model.

Numerical Experiment: One-Parameter Abatement Model

Gaming Mode: You may explore this parameter experiment in our graphical user interface, without coding.



Code Session

- Explore a DICE model with a one-parameter abatement model.

You can find the code of the model implementation in the

package `net.finmath.climate.models.dice`

at <https://github.com/finmath/finmath-lib> (see `src/main/java`).

You can find the code of the numerical experiment in the

class `DICEModelFromGivenParameters`

in the **package** `net.finmath.climateschool.experiments.session1`

at <https://github.com/finmath/climate-school-exercises> (see `src/main/java`).

NUMERICAL EXPERIMENTS

POLICY OPTIMIZATION

Policy Optimization: Optimal Emissions Path

Here, **policy optimization** means choosing the parameters that maximize a given objective function.

- ▶ For complex models, this is usually done numerically: change the parameters, check the new value, try to find the maximum value.
- ▶ For a one-parameter model without local minima, we may use a one-dimensional optimizer such as golden-section search, successively split our search interval, discarding the subinterval that cannot contain the maximum.
- ▶ For multi-parameter models, we can use optimizers such as Levenberg–Marquardt or ADAM.

For the DICE model, we try to find the optimal abatement (and savings-rate) strategy that maximizes total social welfare. In our context, this is called **policy optimization**.

The result is sometimes called *the optimal emissions path*.

NUMERICAL EXPERIMENTS

NUMERICAL EXPERIMENT: OPTIMAL EMISSIONS PATH

Numerical Experiment: Optimal Emission Paths (Calibration) (1)

- ▶ Consider a DICE model with a one-parameter abatement function.
- ▶ Use an optimizer to find the optimal time of reaching maximum abatement in the simple one-parameter DICE model.

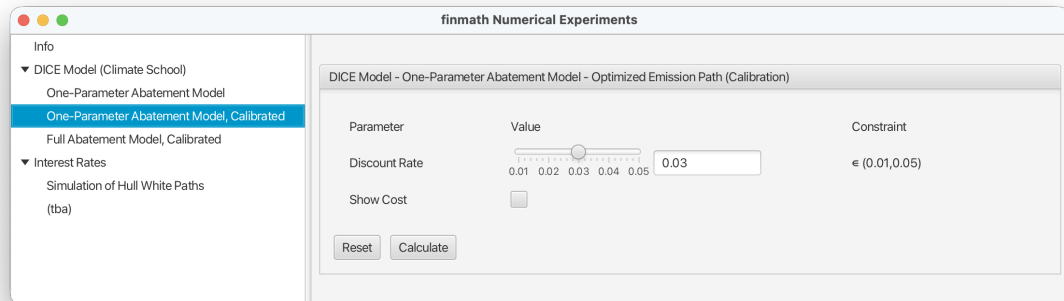
In other words, the abatement-model parameter is the optimal choice that maximizes total welfare: **the optimal emissions path (within this model)**.

This optimal path will depend on the choice of the interest rate.

- ▶ Explore the results for different choices of the interest rate r .
- ▶ What is the dependence of $T^{\mu=1}$ on r ?

Numerical Experiment: Optimal Emission Paths (Calibration) (1)

Gaming Mode: You may explore this parameter experiment in our graphical user interface, without coding.



Code Session

- ▶ Optimize the DICE model with a one-parameter abatement model.
- ▶ Explore the dependence $r \mapsto T^{\mu=1}$.

You can find the code of the numerical experiment in the

class `DICEModelOneParametricCalibration`

in the **package** `net.finmath.climateschool.experiments.session2`

at `https://github.com/finmath/climate-school-exercises` (see `src/main/java`).

Numerical Experiment: Optimal Emission Paths (Calibration) (2)

- ▶ Optimize the DICE model with a full n -parameter abatement function

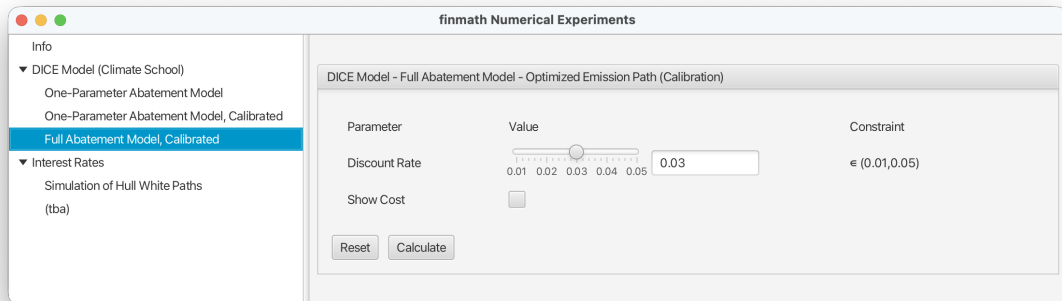
$$\mu(t) = \mu(t_i) \quad \text{for all } t \text{ with } t_i \leq t < t_{i+1},$$

that is, we have many free parameters $\mu(t_0), \mu(t_1), \mu(t_2), \dots$

- ▶ Observe the iterations of the optimizer and interpret the results.

Numerical Experiment: Optimal Emission Paths (Calibration) (2)

Gaming Mode: You may explore this parameter experiment in our graphical user interface, without coding.



Code Session

- ▶ Optimize the DICE model with a full n -parameter abatement function

$$\mu(t) = \mu(t_i) \quad \text{for all } t \text{ with } t_i \leq t < t_{i+1},$$

that is, we have many free parameters $\mu(t_0), \mu(t_1), \mu(t_2), \dots$

- ▶ Observe the iterations of the optimizer and interpret the results.

You can find the code of the numerical experiment in the

class `DICEModelCalibration`

in the **package** `net.finmath.climateschool.experiments.session3`

at <https://github.com/finmath/climate-school-exercises> (see `src/main/java`).

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Social Cost of Carbon

The social cost of carbon asks: **How much additional consumption would compensate for the welfare loss caused by one additional tonne of CO₂?**

For reference, this marginal trade-off is

$$SCC(t) = -1000 \cdot \frac{\partial W}{\partial E(t)} \bigg/ \frac{\partial W}{\partial C_{\text{cons}}(t)},$$

where W is total discounted social welfare. An emission at time t affects welfare in later periods.

- ▶ Dividing the welfare effect of emissions by the welfare effect of consumption expresses the loss in monetary terms.
- ▶ The factor 1000 converts trillion USD per GtCO₂ into USD per tCO₂; the minus sign reports the loss as a positive cost.
- ▶ $SCC(t)$ is a *marginal* cost, not the total cost of all emissions.

SOCIAL COST OF CARBON

NUMERICAL EXPERIMENT: SOCIAL COST OF CARBON

Numerical Experiment: Social Cost of Carbon

- ▶ Calculate the social cost of carbon for a given DICE model.

Remark:

- ▶ In our generalized implementation [1], we estimate the marginal trade-off by applying small shifts and revaluing total welfare:

$$\frac{\partial f(x)}{\partial x} \approx \frac{f(x+h) - f(x)}{h} \quad \text{with } h \text{ small.}$$

The original DICE model obtains the corresponding marginal ratio from GAMS shadow prices.

- ▶ Our `class DICEModel` allows small shifts to be applied to consumption or emissions, followed by revaluation of total welfare.
 - ▶ The parameter `initialConsumptionShift` applies a given shift to consumption.
 - ▶ The parameter `initialEmissionShift` applies a given shift to emissions.
 - ▶ The parameter `isTimeIndexToShift` selects the time index or indices at which the shift is applied.

Code Session

- ▶ Calculate the social cost of carbon for a given DICE model for different values of the discount rate r .

You can find the code of the numerical experiment in the

class `DICEModelSocialCostOfCarbon`

in the **package** `net.finmath.climateschool.experiments.session4`

at `https://github.com/finmath/climate-school-exercises` (see `src/main/java`).

SOCIAL COST OF CARBON

THE SOCIAL COST OF CARBON DOES NOT COVER THE TOTAL CLIMATE COST

- ▶ Assume we would pay $SCC(t)$ for every emission $E(t)$. This corresponds to a total value of

$$\int_0^T SCC(t)E(t) \exp(-rt) dt / 1000 \quad (\text{unit: TrUSD} = 10^{12} \text{ USD}).$$

- ▶ Compare this amount with the total costs associated with climate change:

$$\int_0^T \text{Cost}(t) \exp(-rt) dt \quad (\text{unit: TrUSD} = 10^{12} \text{ USD}).$$

The Social Cost of Carbon Does Not Cover Total Climate Costs

Observation: The two do not match. This is because the SCC is a marginal price in the equilibrium state. The true costs of emissions are far higher. See [2] for details. A numerical experiment based on our model is available.

<https://ssrn.com/abstract=3284722>

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- ▶ From the DICE model, we may extract two types of **climate-related costs**: abatement costs and damage costs.
- ▶ We use one burden metric: these costs relative to contemporaneous GDP, i.e., **costs as a share of GDP**.
- ▶ Under this metric, the optimized path in this implementation produces an uneven intergenerational burden:
 - ▶ The climate-related cost share is larger for future generations than for current generations; policy optimization increases this effect in this experiment.

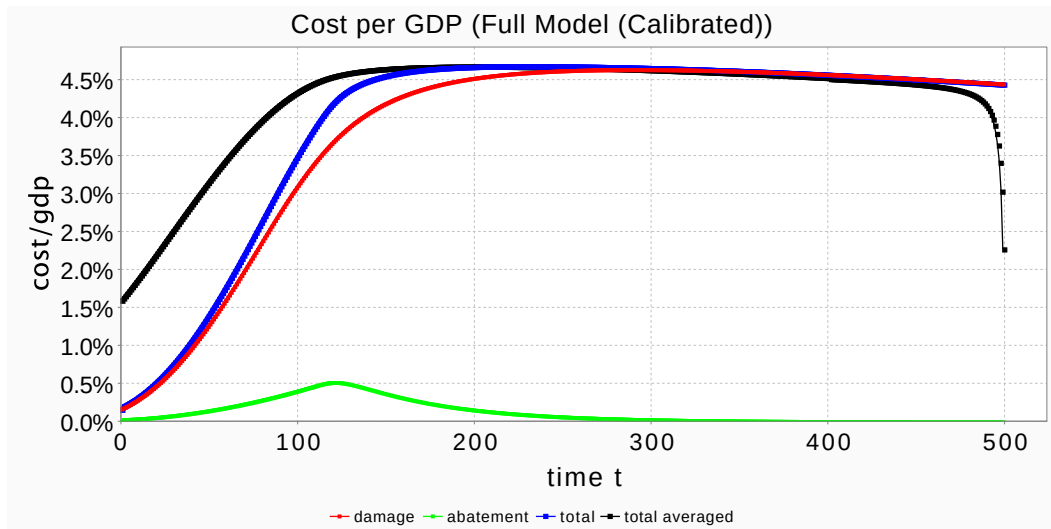


Figure 4: Costs as percentages of GDP in the optimized model. Damage costs (red), abatement costs (green), and their sum (blue). Black shows a forward running average over a 100-year window.

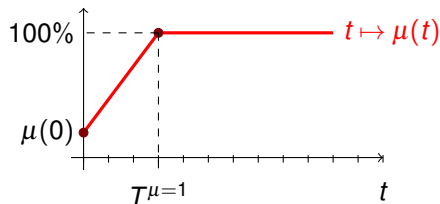
INTERGENERATIONAL EQUITY IN INTEGRATED ASSESSMENT MODELS

NUMERICAL EXPERIMENT: DISTRIBUTION OF COST IN THE ONE-PARAMETER ABATEMENT MODEL

Numerical Experiment: Distribution of Cost in the One-Parameter Abatement Model

- ▶ Instantiate an object of `class` `DICEModel`.
 - ▶ Use a constant savings rate $t \mapsto 0.26$.
 - ▶ Use a simple one-parameter abatement model, with the single parameter $T^{\mu=1}$: linear from 0.03 at $t = 0$ to 1.0 at $t = T^{\mu=1}$, then constant at 1.0.

$$\mu(t) = \min \left(\mu(0) + \frac{1 - \mu(0)}{T^{\mu=1}} t, 1.0 \right),$$



- ▶ Analyse the distribution of costs in the one-parameter abatement model.

Note: This is the same numerical experiment as the first one, but with a different output.

Numerical Experiment: Distribution of Cost in the One-Parameter Abatement Model

Gaming Mode: Explore **the distribution of costs** in the one-parameter model.

The screenshot shows a software interface titled "finmath Numerical Experiments". On the left is a sidebar menu with the following items: "Info", "DICE Model (Climate School)" (expanded), "One-Parameter Abatement Model" (selected), "One-Parameter Abatement Model, Calibrated", "Full Abatement Model, Calibrated", "Interest Rates" (expanded), "Simulation of Hull White Paths (tba)", and "Simulation of Hull White Paths (tba)". The main panel is titled "DICE Model - One-Parameter Abatement Model - NOT CALIBRATED". It contains a table with three columns: "Parameter", "Value", and "Constraint".

Parameter	Value	Constraint
Discount Rate	0.03	€ (0.01,0.05)
Abatement Max Time	50	€ (10,200)

Below the table, there is a checkbox labeled "Show Cost" which is checked. A red arrow points to this checkbox. At the bottom of the panel are two buttons: "Reset" and "Calculate".

Why are abatement costs so small, but damage costs so large?

Let $C(t)$ denote the total cost observed in time t . It is the sum of the abatement cost $C_A(t)$ and the damage cost $C_D(t)$. Let's investigate the objective function:

We consider a simple abatement model with a single parameter $T^{\mu=1}$. The calibration will choose the parameter $T^{\mu=1}$ that maximizes $\int_0^T V(t) \frac{N(0)}{N(t)} dt$. Hence,

$$\begin{aligned} \frac{d}{dT^{\mu=1}} \int_0^T V(t) \frac{N(0)}{N(t)} dt &= \int_0^T \frac{dV(t)}{dC(t)} \frac{dC(t)}{dT^{\mu=1}} \frac{N(0)}{N(t)} dt \\ &= \int_0^T \frac{dV(t)}{dC(t)} \left(\frac{dC_D(t)}{dT^{\mu=1}} + \frac{dC_A(t)}{dT^{\mu=1}} \right) \frac{N(0)}{N(t)} dt \stackrel{!}{=} 0 \end{aligned} \quad (1)$$

Note that $\frac{dC_D(t)}{dT^{\mu=1}} > 0$ and $\frac{dC_A(t)}{dT^{\mu=1}} < 0$.

Why are abatement costs so small, but damage costs so large?

Our numerical experiment fixes savings and varies only one policy parameter: the time $T^{\mu=1}$ at which modelled industrial emissions are fully abated. The original DICE model instead optimizes complete paths for abatement and savings.

At the optimum, a small delay in reaching full abatement cannot improve welfare. Its effects approximately balance:

Welfare effect of a small delay

Savings from lower near-term abatement cost + losses from greater climate damage + indirect effects on investment and output ≈ 0 .

The calibration balances **marginal welfare gains and losses**, not the total levels of abatement cost and damage cost.

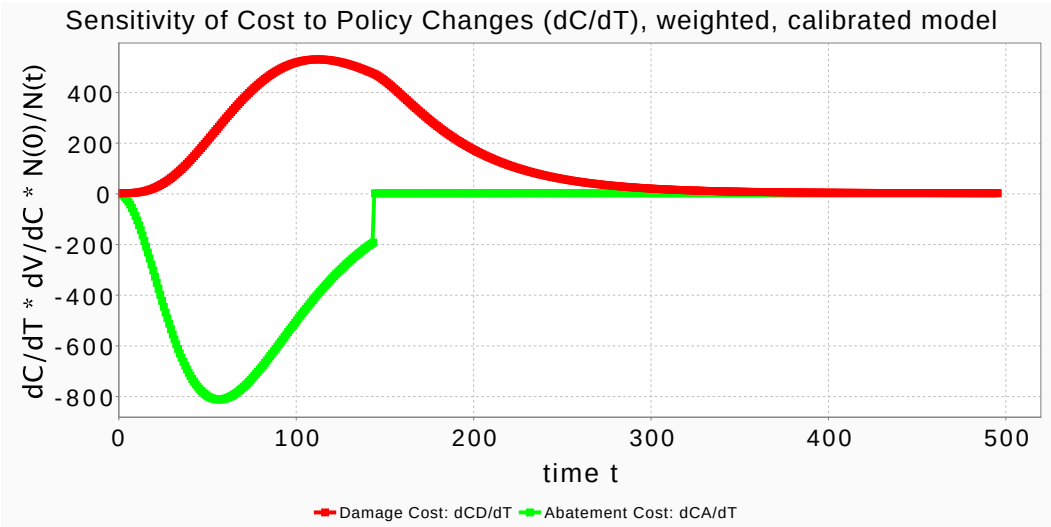


Figure 5: Weighted direct cost sensitivities for a small delay in reaching full abatement; the indirect output effect is omitted.

Figure 5 reflects several effects:

- ▶ Abatement costs occur earlier (immediately) than damage costs.
- ▶ Once both compared policies have reached 100% abatement, contemporaneous abatement cost is no longer directly sensitive to $T^{\mu=1}$. Damage may nevertheless remain sensitive because of earlier emissions.
- ▶ The complete welfare sensitivity also contains the indirect effect on future capital and output.

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Convexity

- ▶ The following revisits an effect from probability theory that may already be familiar.
- ▶ In mathematical finance, it is called the *convexity effect*.

Expectation of a Linear Function of a Random Variable

- ▶ Consider the mapping $x \mapsto y = f(x) := 6 \cdot x - 4$.
- ▶ Assume that X is random with expectation (average) μ , i.e., $\mu = E(X) = 2$.
- ▶ What is the expectation of the mapped values Y (that is, of $Y = f(X) = 6 \cdot X - 4$)?
- ▶ The answer is $6 \cdot \mu - 4 = 8$.
- ▶ **Let us illustrate this!** $\rightarrow$ *Computer Experiment*.

Observation

Note that the result above depends only on the expectation of X , that is, on μ .

The variance of X does not matter.

However, this is not always the case.

Expectation of a **Nonlinear** Function of a Random Variable

- ▶ Consider the mapping $x \mapsto y = f(x) := x^2$.
- ▶ Assume that X is random with expectation (average) μ , i.e., $\mu = E(X) = 2$.
- ▶ What is the expectation of the mapped values Y (that is, of $Y = f(X) = X^2$)?
- ▶ The answer depends on the variance!
- ▶ **Let us illustrate this!** → *Computer Experiment*.

Observation

Note that the expectation $E(Y)$ increases as the variance of X increases.

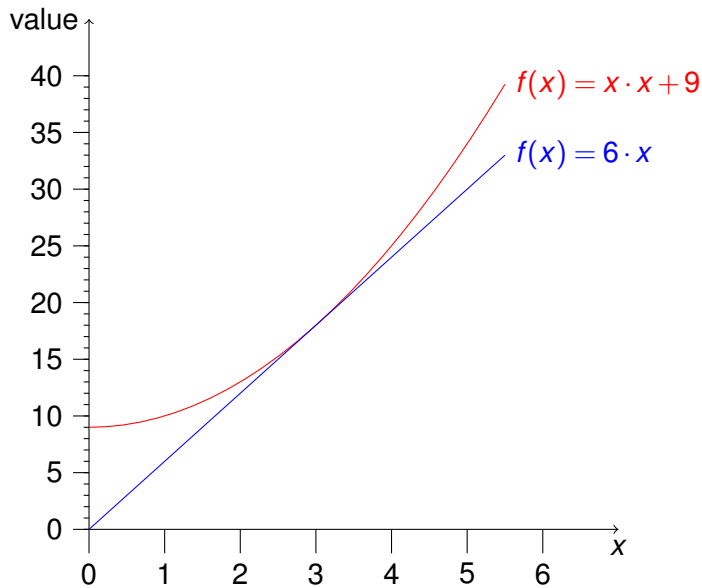
Variance (or standard deviation) measures dispersion and therefore captures one aspect of *risk*: the possibility that X deviates from its expectation μ .

For the convex function $f(x) = x^2$, more variance at a fixed mean increases the expectation of the result.

Hence, replacing a random input by its mean can understate expected outcomes for convex mappings.

We will observe this: uncertainty in interest rates increases intergenerational inequality.

Convexity



Code Session: Convexity

- ▶ Illustrate the effect of convexity on the expectation of a function of a random variable.

You can find the code of the numerical experiment in the

class `ConvexityExperiments`

in the **package** `net.finmath.climateschool.experiments.session5`

at `https://github.com/finmath/climate-school-exercises` (see `src/main/java`).

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Stochastic Interest Rates

- ▶ Interest rates change over time.
- ▶ We do not know how they change in the future—there are different possible values that occur with certain probabilities.
- ▶ In mathematical finance, there is a rich family of stochastic interest-rate models.
- ▶ In computational finance, a popular method is Monte Carlo simulation.
- ▶ We simulate thousands of scenarios ω for the future path $t \mapsto r(t, \omega)$.
- ▶ $r(t)$ is now a random variable. Here, $t \mapsto r(t)$ is called a stochastic process.
- ▶ We can then analyse the resulting distribution, including its expectation and quantiles.

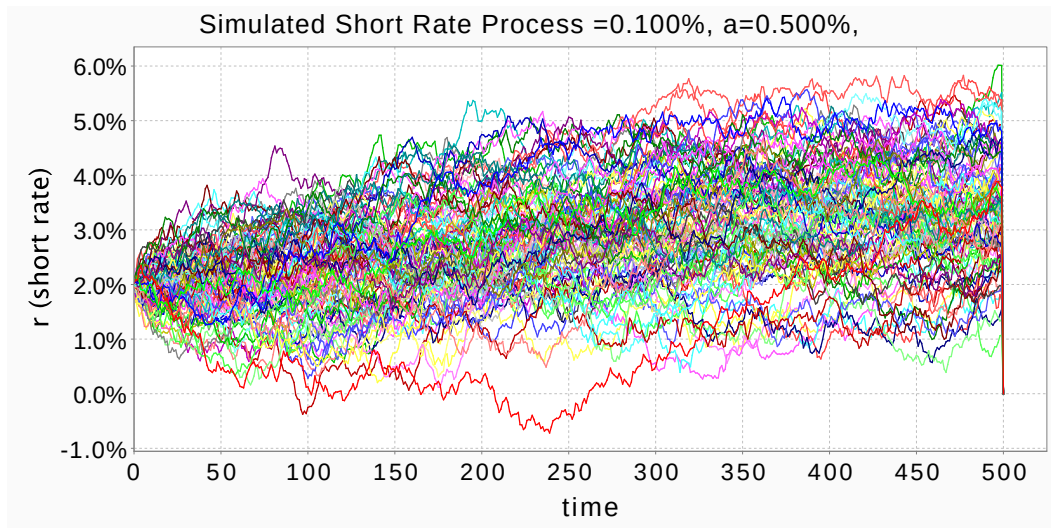


Figure 6: A simulation of the interest rate r (short rate). This is a Hull–White model with parameters $\sigma = 0.1\%$ (volatility) and $a = 0.5\%$ (mean reversion speed).

Code Session: Stochastic Interest Rates

- ▶ Generate and plot sample paths of a classical Hull–White model.

You can find the code of the numerical experiment in the

class `StochasticRatesExperiment`

in the **package** `net.finmath.climateschool.experiments.session6`

at <https://github.com/finmath/climate-school-exercises> (see `src/main/java`).

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IAM with Stochastic Interest Rates

- ▶ Given that interest rates have a strong impact on the optimal abatement policy, a change in the interest-rate curve should lead to a change in the abatement policy (see Figure 3).
- ▶ Hence, we introduce a **stochastic abatement model** that responds to changes in the interest-rate level.
- ▶ Through this, all model quantities, including abatement costs and damage costs, become stochastic.

Stochastic Abatement Model

We consider a very simple three-parameter stochastic model in which the abatement speed can be reduced or increased as a function of the current interest-rate level r (short rate).

The model is

$$\mu(t, \omega) = \min(\mu_0 + (a_0 + a_1 \cdot r(t, \omega)) \cdot t, 1.0). \quad (2)$$

The parameter a_0 is the speed at which we approach 100 % abatement, but with $a_1 \neq 0$ this speed depends on the interest rate level. Recall Figure 3.

From Figure 3 we expect that a_1 is negative: if the interest-rate level is higher, abatement will be slower.

Stochastic Abatement Model

With a stochastic abatement model, all quantities become stochastic: emissions, atmospheric carbon, damage, and costs.

The stochastic abatement model also lets us analyse how *risk* is distributed across generations.

The following figure shows expected abatement and damage costs together with their simulated uncertainty bands, following the optimized strategy within this model.

Surprise: In this experiment, if the abatement policy adapts to changing interest-rate levels, the uneven intergenerational burden becomes *larger*.

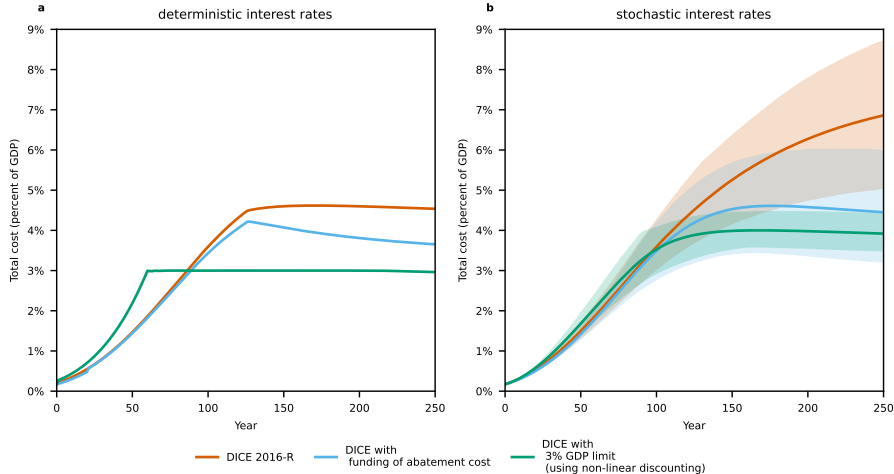


Figure 7: Costs as shares of GDP with deterministic interest rates (left) and stochastic interest rates with an adaptive abatement policy (right). Uncertainty increases the expected level (line) and introduces an uncertainty corridor.

Stochastic Abatement Model

Insights

Future generations are exposed to higher expected costs and significant **risk** in damage costs.

Within this experiment, repeated adaptation of the abatement strategy to interest-rate changes increases the risk of high damage costs.

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Extensions that Improve Intergenerational Equity

In our recent work [5], we show that two simple model extensions improve intergenerational equity:

- ▶ Allowing funding of abatement costs: abatement costs are funded by a loan.
- ▶ Introducing nonlinear discounting of damage costs: large damages receive a larger weight (a larger discount factor).

Both extensions have an economic interpretation and motivation.

EXTENSIONS THAT IMPROVE INTERGENERATIONAL EQUITY

ADDING FUNDING TO INTEGRATED ASSESSMENT MODELS

Funding of Abatement Costs

We add the following modification to the model:

- ▶ Funding of abatement costs: we allow the abatement costs at time t to be financed by a loan, repaid at time $t + \Delta T$ with a market forward rate $FR(t, t + \Delta T; t)$.

In a risk-neutral valuation model, this modification would have no effect, given that the forward rate is derived from the discount factor. For the DICE model, we see an effect. The reason is the change in the sensitivity of the utility.

⇒ Allowing funding

- ▶ improves cost-related intergenerational equity under our metric (Figure 8),
- ▶ reduces cost risk in a stochastic model (Figure 9), and
- ▶ induces earlier abatement in the optimized policy (Figure 10).

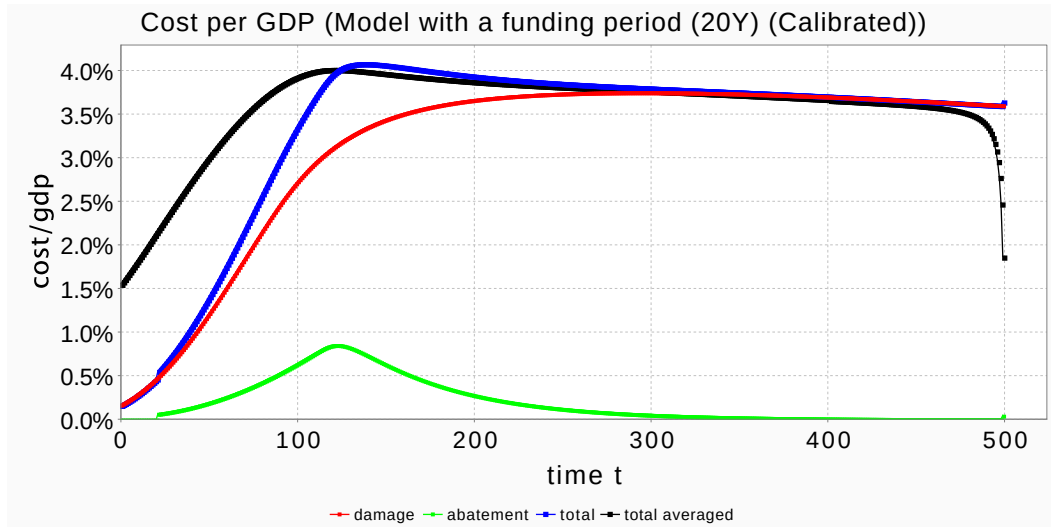


Figure 8: Costs as percentages of GDP in the optimized model with a 20-year funding period for abatement. Damage costs (red), abatement costs (green), and their sum (blue). Black shows a forward running average over a 100-year window.

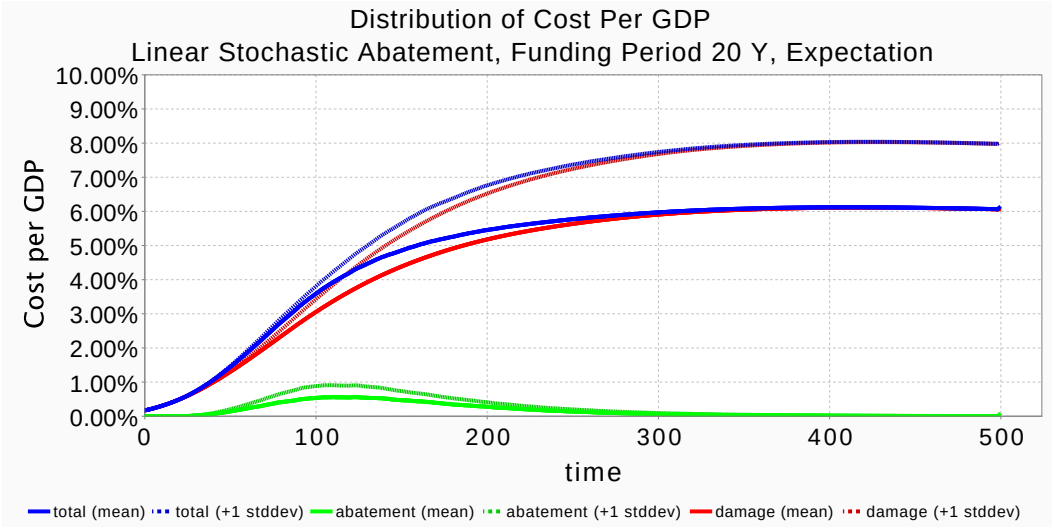


Figure 9: Distribution of undiscounted costs as a share of GDP using stochastic interest rates, the abatement model (2), and funding of abatement.

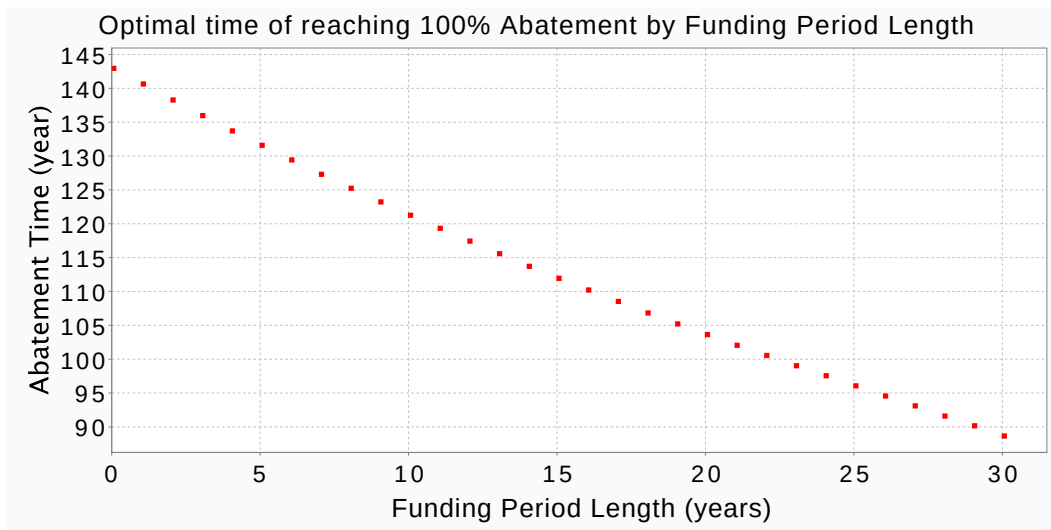


Figure 10: The calibrated (optimal) parameter $T^{\mu=1}$ for models using different funding periods.

Within this model, a 20-year funding period for abatement costs shifts the preferred abatement time roughly 40 years earlier.

EXTENSIONS THAT IMPROVE INTERGENERATIONAL EQUITY

ADDING NONLINEAR DISCOUNTING TO INTEGRATED ASSESSMENT MODELS

We add the following modification to the model:

- ▶ Nonlinear discounting: larger costs receive a larger discount factor (modelling that large projects or damages are usually over budget).

Adding nonlinear discounting of costs is a simple way to limit costs. Applying a larger weight (discount factor) to large costs than to small costs acts like a penalty. Policy optimization will favour a more even distribution of costs among generations.

We can apply a soft penalty by assigning greater weight to costs that exceed 3% of GDP. The optimized solution then nearly respects this threshold:

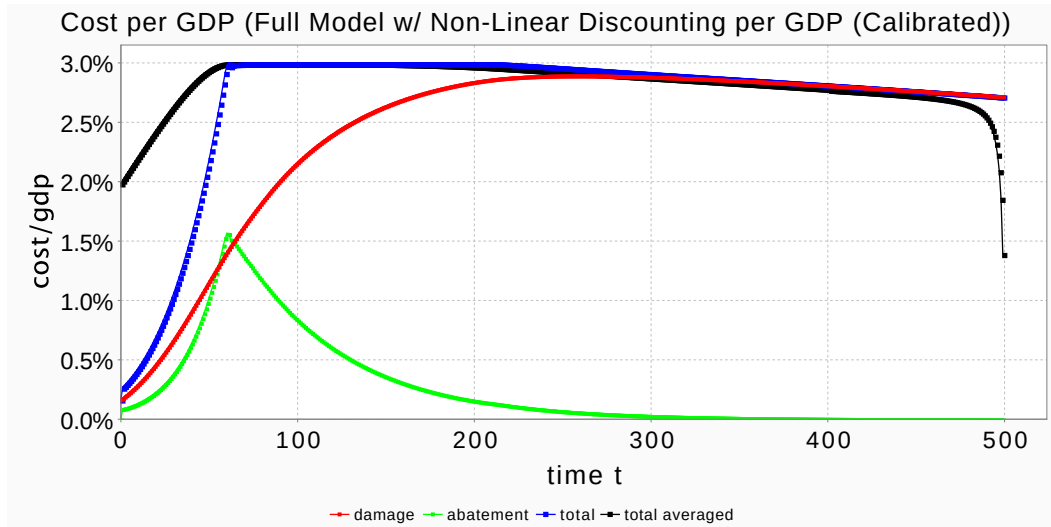


Figure 11: The temporal distribution of total costs as a share of GDP under strong nonlinear discounting.

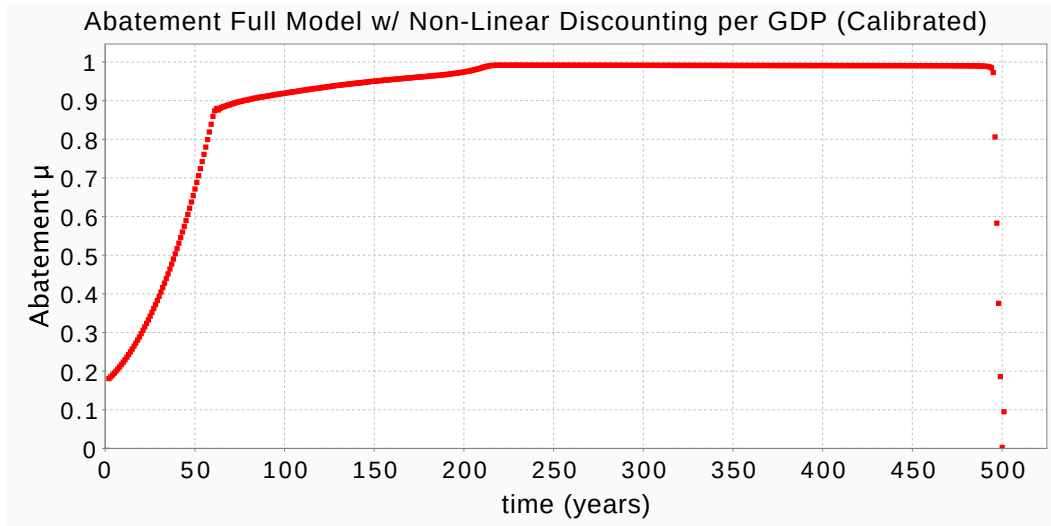


Figure 12: The optimal abatement function $t \mapsto \mu(t)$ under nonlinear discounting applied to costs above 3% of GDP.

Improving Intergenerational Equity

In other words, to keep the cost to future generations below 3% of GDP, we should ramp up abatement rapidly—and contribute our share.

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- ▶ In this implementation, the optimized DICE path produces a rising climate-related cost share across generations.
- ▶ This reflects the combined effects of
 - ▶ policy optimization balancing marginal changes in costs, and
 - ▶ the sensitivity of utility to discounting.
- ▶ In the adaptive-policy experiment, stochastic interest rates expose future generations to additional cost risk and amplify this uneven burden.
- ▶ Allowing abatement costs to be funded over time shifts the optimized strategy towards earlier abatement and a more even burden.
- ▶ Nonlinear weighting of large costs likewise produces a more even cost burden in the optimized climate-mitigation pathway.

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SOURCE CODE

Implementation (Open Source)

- ▶ You may explore the code of the DICE model, our extensions and all numerical experiments by checking out our open source repositories.

You can find the code of the classical DICE model implementation in the

package `net.finmath.climate.models.dice`

at <https://github.com/finmath/finmath-lib> (see `src/main/java`).

You can find our stochastic DICE model and extensions

at <https://gitlab.com/finmath/finmath-climate-nonlinear>.

Source Code

You can find the code of the numerical experiments

at `https://github.com/finmath/climate-school-exercises`.

You can also find code with numerical experiments related to the classical DICE model in

class `DICEModelExperiment`

in the **package** `net.finmath.experiments.dice`

at `https://github.com/finmath/finmath-experiments` (see `src/main/java`).

SOURCE CODE AND REFERENCES

REFERENCES

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